

49. MultiMediaCard Interface (MMCIF)

49.1 Overview

This MCU incorporates a MultiMediaCard interface (MMCIF) that conforms to the JEDEC Standard JESD84-A441. Table 49.1 lists the specifications of the MMCIF.

Table 49.1 MMCIF Specifications

Item	Function
MMC bus interface	Transfer bus width selectable from 1, 4, or 8 bits
Transfer modes	• Backward compatible mode or high-speed mode selectable • Supports the single data rate • Supports block transfer (stream read and stream write operations are not supported)
Clocks	The MMCIF clock is generated by dividing peripheral module clock B (PCLKB) by n , where $n = 2, 4, 8, 16, 32, 64, 128, 256, 512$, or $1,024$.
Boot operation mode	The MMCIF clock frequency is switchable during boot operation (alternative boot operation is not supported).
MMCIF buffers	512 bytes $\times$ 2
Interrupt sources	• Normal operation interrupt (ACCIO) • Error/timeout interrupt (ERRIO) • MMC detection interrupt (CDETIO) • MMCIF buffer access interrupt (MBFAI)
DMA transfer requests	• MBFAI interrupt can be used to trigger the DMAC and DTC • DMAC or DTC can be used to read and write to the MMCIF buffer
Other functions	• MMC detection • Supports background operations • Supports the high priority interrupt (HPI) during CMD6, CMD24, CMD25 (pre-defined), and CMD38 command sequences

Figure 49.1 shows a Block Diagram of the MMCIF.

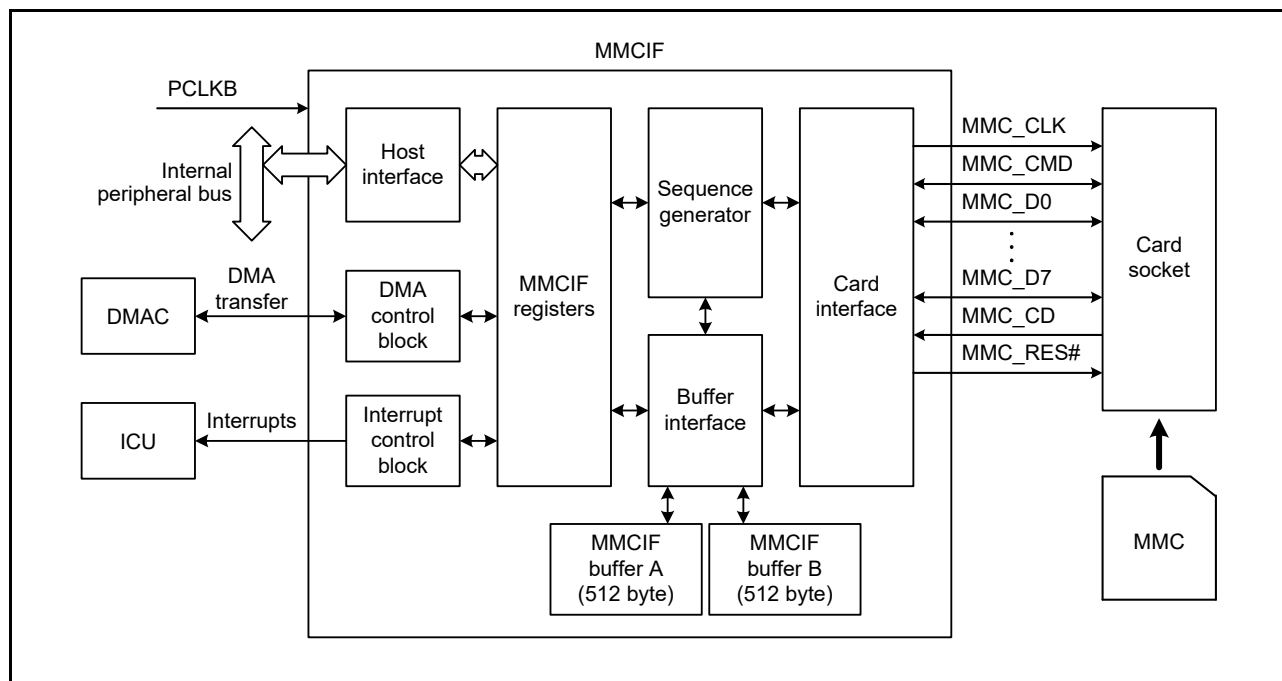


Figure 49.1 Block Diagram of the MMCIF

Table 49.2 lists the MMCIF I/O Pins.

Table 49.2 MMCIF I/O Pins

Pin Name	I/O	Description
MMC_CLK	Output	MMCIF clock
MMC_CMD	I/O	Command output and response input
MMC_D0 to MMC_D7	I/O	Transfer data (DAT0 to DAT7)
MMC_CD	Input	MMC detection *1
MMC_RES#	Output	MMC reset *2

Note 1. Check the specifications of the card socket to be used before connecting the pins to the MMC.

Note 2. Similar reset control is possible with a general purpose port.

49.2.1 Command Setting Register (CECMDSET)

Figure 10-10 shows the bit fields of Register 0x00000000 (R0). The register is 32 bits wide, divided into two 16-bit halves. The top half (bits 31-16) contains fields: bit 31 (—), bits 30-29 (BOOT), bits 28-24 (CMD[5:0]), bits 23-22 (RTYP[1:0]), bit 21 (RBSY), bit 20 (—), bits 19-18 (WDAT, DWEN), bit 17 (CMLTE), and bit 16 (CMD12EN). The bottom half (bits 15-0) contains fields: bits 15-14 (RIDXC[1:0]), bits 13-12 (RCRC7C[1:0]), bit 11 (—), bits 10-9 (CRC16C, BOOTACK), bit 8 (CRCSTE), bit 7 (TBIT), bit 6 (OPDM), bit 5 (—), bit 4 (—), bit 3 (SBIT), bit 2 (—), and bits 1-0 (DATW[1:0]).

Bit	Symbol	Bit Name	Description	R/W
b1, b0	DATW[1:0]	Data Bus Width Select *1	b1 b0 0 0 : 1-bit data bus width 0 1 : 4-bit data bus width 1 0 : 8-bit data bus width 1 1 : Do not set this value.	R/W
b2	—	Reserved	This bit is 0 when read. Set it to 0 when writing.	R/W
b3	SBIT	Read Data Start Bit Detection Setting *2	0: When the MMC_Dn pins corresponding to the data bus width set in the DATW[1:0] bits are all low, this bit is detected as a start bit (n = 0, 0 to 3, or 0 to 7) 1: When the MMC_D0 pin is low, this bit is detected as a start bit	R/W
b5, b4	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b6	OPDM	MMC_CMD Output Select	0: Normal output from the MMC_CMD pin 1: Open-drain output from the MMC_CMD pin	R/W
b7	TBIT	Transmission Bit Setting	0: Transmission bit is set to 1 1: Transmission bit is set to 0	R/W
b8	CRCSTE	CRC Status Token Reception *3	0: CRC status token received 1: CRC status token not received (set to 1 when CMD19 is issued)	R/W
b9	BOOTACK	Boot Acknowledge Reception	0: Boot acknowledge not received 1: Boot acknowledge received	R/W
b10	CRC16C	CRC16 Check During Reception *2	0: Check CRC16 in the transfer data 1: Do not check CRC16 in the transfer data (set to 1 when CMD14 is issued)	R/W
b11	—	Reserved	This bit is 0 when read. Set it to 0 when writing.	R/W
b13, b12	RCRC7C[1:0]	Response CRC7 Check	b13 b12 0 0 : Performs an error check on CRC7 field value (set the RTYP[1:0] bits to 01b) 0 1 : Confirms that the field values for the response check bits are all 1 (set the RTYP[1:0] bits to 01b) 1 0 : Performs an error check on internal CRC7 field value (set the RTYP[1:0] bits to 10b) 1 1 : Error check not performed on response's CRC7 field value	R/W
b15, b14	RIDXC[1:0]	Response Index Check	b15 b14 0 0 : The command index field value checks to see if the command matches the response. 0 1 : Checks to see if the check bits in the response are all 1 1 0 : Response's command index field value is not checked 1 1 : Do not set this value.	R/W
b16	CMD12EN	CMD12 Automatic Issue Setting	0: CMD12 is not automatically issued during a multiple block transfer 1: CMD12 is automatically issued during a multiple block transfer	R/W
b17	CMLTE	Transfer Type Select *1	0: Single block transfer 1: Multiple block transfer	R/W
b18	DWEN	MMC Access Select *1	0: Read data from the MMC 1: Write data to the MMC	R/W
b19	WDAT	Data Presence Determination	0: No data present in the MMC 1: Data present in the MMC	R/W
b20	—	Reserved	This bit is 0 when read. Set it to 0 when writing.	R/W

Bit	Symbol	Bit Name	Description	R/W
b21	RBSY	MMC Response	0: No response busy 1: Response busy is included (R1b)	R/W
b23, b22	RTYP[1:0]	Response Type Select	b23 b22 0 0 : No response 0 1 : 6-byte response (R1, R1b, R3, R4, R5) 1 0 : 17-byte response (R2) 1 1 : Do not set this value.	R/W
b29 to b24	CMD[5:0]	Command Index	Set the command index field value.	R/W
b30	BOOT	Boot Operations	0: Command sequence when not performing boot operations 1: Command sequence when performing boot operations	R/W
b31	—	Reserved	This bit is 0 when read. Set it to 0 when writing.	R/W

Note 1. The DATW[1:0], CMLTE, and DWEN bit values are valid when the WDAT bit is 1.

Note 2. The SBIT and CRC16C bit values are valid when the WDAT bit is 1 and the DWEN bit is 0.

Note 3. The CRCSTE bit value is valid when bits WDAT and DWEN are both 1.

The CECMDSET register defines the command sequence. The command sequence starts when a value is written to the CECMDSET register. The CECMDSET register cannot be write accessed when the CEHOSTSTS1.CMDSEQ bit is 1.

Table 49.3 lists the setting values for the CECMDSET register.

CMD12EN Bit (CMD12 Automatic Issue Setting)

Set this bit to select whether CMD12 is automatically issued or not during a multiple block transfer. When the CMD12EN bit is 1 and CMD12 automatic issuance is enabled, set the RBSY bit to 0 and set the CEBLOCKSET.BLKSIZ[15:0] bits to 200h (512 bytes). For details on CMD12 automatic issuance, refer to section 49.3.4, Automatically Issuing CMD12.

RBSY Bit (MMC Response)

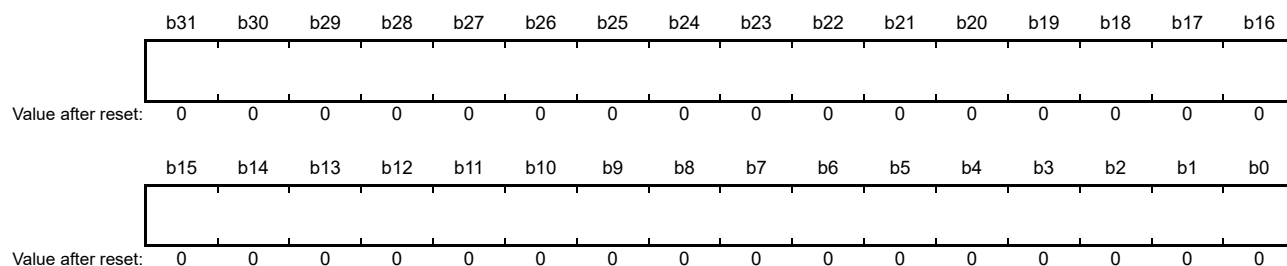
Set this bit to enable or disable the busy signal when receiving a response.

Table 49.3 Setting Values for the CECMDSET Register

Command	Response	CECMDSET Register Setting Value																						Remark
		—	BOOT	CMD[5:0]	RTYP[1:0]	RBSY	—	WDAT	DWEN	CMLTE	CMD12EN	RIDXC[1:0]	RCRC7C[1:0]	—	CRC16C	BOOTACK	CRCSTE	TBIT	OPDM	—	—	SBIT	—	
CMD0	—	0	0	000000	00	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD1	R3	0	0	000001	01	0	0	0	0	0	0	01	01	0	0	0	0	0	0	0	0	0	00	
CMD2	R2	0	0	000010	10	0	0	0	0	0	0	01	10	0	0	0	0	0	0	0	0	0	00	
CMD3	R1	0	0	000011	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD4	—	0	0	000100	00	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD5	R1b	0	0	000101	01	1	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD6	R1	0	0	000110	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	BGO
	R1b	0	0	000110	01	1	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD7	R1	0	0	000111	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
	R1b	0	0	000111	01	1	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD8	R1	0	0	001000	01	0	0	1	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD9	R2	0	0	001001	10	0	0	0	0	0	0	01	10	0	0	0	0	0	0	0	0	0	00	
CMD10	R2	0	0	001010	10	0	0	0	0	0	0	01	10	0	0	0	0	0	0	0	0	0	00	
CMD12	R1	0	0	001100	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
	R1b	0	0	001100	01	1	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD13	R1	0	0	001101	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
	R1b	0	0	001101	01	1	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	HPI
CMD14	R1	0	0	001110	01	0	0	1	0	0	0	00	00	0	1	0	0	0	0	0	0	1	00	**
CMD15	—	0	0	001111	00	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD16	R1	0	0	010000	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD17	R1	0	0	010001	01	0	0	1	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD18	R1	0	0	010010	01	0	0	1	0	1	0	00	00	0	0	0	0	0	0	0	0	0	00	**
	R1	0	0	010010	01	0	0	1	0	1	1	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD19	R1	0	0	010011	01	0	0	1	1	0	0	00	00	0	0	0	1	0	0	0	0	0	00	**
CMD23	R1	0	0	010111	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD24	R1	0	0	011000	01	0	0	1	1	0	0	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD25	R1	0	0	011001	01	0	0	1	1	1	0	00	00	0	0	0	0	0	0	0	0	0	00	**
	R1	0	0	011001	01	0	0	1	1	1	1	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD26	R1	0	0	011010	01	0	0	1	1	0	0	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD27	R1	0	0	011011	01	0	0	1	1	0	0	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD28	R1b	0	0	011100	01	1	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD29	R1b	0	0	011101	01	1	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD30	R1	0	0	011110	01	0	0	1	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD31	R1	0	0	011111	01	0	0	1	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD35	R1	0	0	100011	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD36	R1	0	0	100100	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD38	R1b	0	0	100110	01	1	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD39	R4	0	0	100111	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD40	R5	0	0	101000	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	Send CMD
	R5	0	0	101000	01	0	0	0	0	0	0	00	00	0	0	0	0	1	1	0	0	0	00	Send RSP
CMD42	R1	0	0	101010	01	0	0	1	1	0	0	00	00	0	0	0	0	0	0	0	0	0	00	**
CMD55	R1	0	0	110111	01	0	0	0	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	
CMD56	R1	0	0	111000	01	0	0	1	0	0	0	00	00	0	0	0	0	0	0	0	0	0	00	Read
	R1	0	0	111000	01	0	0	1	1	0	0	00	00	0	0	0	0	0	0	0	0	0	00	Write
Boot operation		0	1	000000	00	0	0	1	0	1	0	00	00	0	0	*	0	0	0	0	0	0	00	**

49.2.2 Argument Register (CEARG)

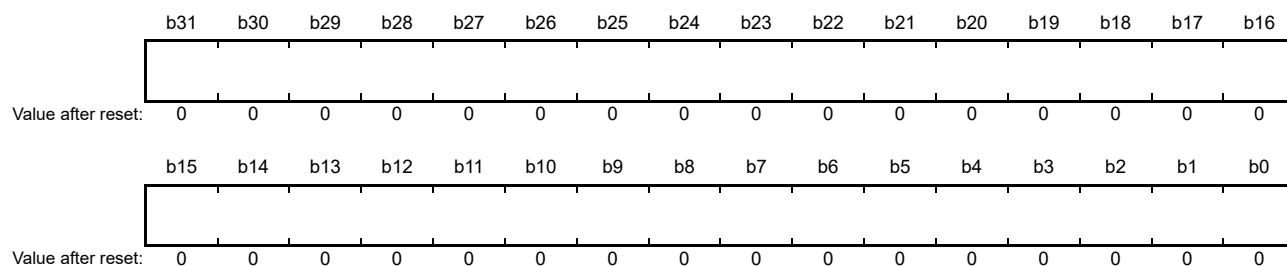
Address(es): 0008 8508h



The CEARG register defines the argument field value for the command issued. Set a value to the CEARG register before starting the command sequence. The argument field value for an automatically issued CMD12 should be set in the CEARGCMD12 register.

49.2.3 Automatically Issued CMD12 Argument Register (CEARGCMD12)

Address(es): 0008 850Ch



The CEARGCMD12 register defines the argument field value of an automatically issued CMD12. This register is only accessible when CMD12 is automatically issued during a multiple block transfer. Set a value to the CEARGCMD12 register before starting the command sequence. For details on CMD12 automatic issuance, refer to [section 49.3.4, Automatically Issuing CMD12](#).

49.2.4 Command Control Register (CECMDCTRL)

Address(es): 0008 8510h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BREAK
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	BREAK	Command Sequence Force Stop	Writing 1 to this bit stops the command sequence. This bit will not automatically revert to 0, so after setting it to 1, set it back to 0.	R/W
b31 to b1	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W

BREAK Bit (Command Sequence Force Stop)

To stop the command sequence, when the BREAK bit is 0, set it to 1 and then to 0 in succession. Then confirm that the CEHOSTSTS1.CMDSEQ flag is 0, set the CEVERSION.SWRST bit to 1, and perform an MMCIF software reset. When an MMCIF software reset is performed, registers associated with the MMCIF module become their value after reset values, so initial settings are required.

49.2.5 Transfer Block Setting Register (CEBLOCKSET)

Address(es): 0008 8514h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	BLKCNT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	BLKSIZ[15:0]															
Value after reset:	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	BLKSIZ[15:0]	Transfer Block Size	These bits define the transfer block size. When performing a single block transfer, values equivalent to 1 to 512 (bytes) can be set. When performing a multiple block transfer, set a value equivalent to 512 (bytes).	R/W
b31 to b16	BLKCNT[15:0]	Number of Transfer Blocks *1	These bits define the number of transfer blocks.	R/W

Note 1. These bits are accessible during a multiple block transfer.

The CEBLOCKSET register defines the block size of the data to be transferred, and the number of blocks. Set a value to the CEBLOCKSET register before starting the command sequence.

49.2.6 Clock Control Register (CECLKCTRL)

Address(es): 0008 8518h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	MMCB USBSY	—	—	—	—	—	—	CLKEN	—	—	—	—	CLKDIV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	SRSPTO[1:0]		SRBSYTO[3:0]			SRWDTO[3:0]			—			—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description		R/W
b3 to b0	—	Reserved	These bits are 0 when read. Set them to 0 when writing.		R/W
b7 to b4	SRWDTO[3:0]	Write Data Timeout/Read Data Timeout Setting	b7 b4 0 0 0 0: MMCIF × 2 ¹⁴ 0 0 0 1: MMCIF × 2 ¹⁵ 0 0 1 0: MMCIF × 2 ¹⁶ 0 0 1 1: MMCIF × 2 ¹⁷ 0 1 0 0: MMCIF × 2 ¹⁸ 0 1 0 1: MMCIF × 2 ¹⁹ 0 1 1 0: MMCIF × 2 ²⁰ 0 1 1 1: MMCIF × 2 ²¹	b7 b4 1 0 0 0: MMCIF × 2 ²² 1 0 0 1: MMCIF × 2 ²³ 1 0 1 0: MMCIF × 2 ²⁴ 1 0 1 1: MMCIF × 2 ²⁵ 1 1 0 0: MMCIF × 2 ²⁶ 1 1 0 1: MMCIF × 2 ²⁷ 1 1 1 0: MMCIF × 2 ²⁸ 1 1 1 1: MMCIF × 2 ²⁹	R/W
b11 to b8	SRBSYTO[3:0]	Response Busy Timeout Setting	b11 b8 0 0 0 0: MMCIF × 2 ¹⁴ 0 0 0 1: MMCIF × 2 ¹⁵ 0 0 1 0: MMCIF × 2 ¹⁶ 0 0 1 1: MMCIF × 2 ¹⁷ 0 1 0 0: MMCIF × 2 ¹⁸ 0 1 0 1: MMCIF × 2 ¹⁹ 0 1 1 0: MMCIF × 2 ²⁰ 0 1 1 1: MMCIF × 2 ²¹	b11 b8 1 0 0 0: MMCIF × 2 ²² 1 0 0 1: MMCIF × 2 ²³ 1 0 1 0: MMCIF × 2 ²⁴ 1 0 1 1: MMCIF × 2 ²⁵ 1 1 0 0: MMCIF × 2 ²⁶ 1 1 0 1: MMCIF × 2 ²⁷ 1 1 1 0: MMCIF × 2 ²⁸ 1 1 1 1: MMCIF × 2 ²⁹	R/W
b13, b12	SRSPTO[1:0]	Response Timeout Setting	b13 b12 0 0 : MMCIF clock × 64 0 1 : MMCIF clock × 128	b13 b12 1 0 : MMCIF clock × 256 1 1 : Do not set this value.	R/W
b15, b14	—	Reserved	These bits are 0 when read. Set them to 0 when writing.		R/W
b19 to b16	CLKDIV[3:0]	MMCIF Clock Frequency Select *1	b19 b16 0 0 0 0: PCLKB/2 0 0 0 1: PCLKB/2 ² 0 0 1 0: PCLKB/2 ³ 0 0 1 1: PCLKB/2 ⁴ 0 1 0 0: PCLKB/2 ⁵	b19 b16 0 1 0 1: PCLKB/2 ⁶ 0 1 1 0: PCLKB/2 ⁷ 0 1 1 1: PCLKB/2 ⁸ 1 0 0 0: PCLKB/2 ⁹ 1 0 0 1: PCLKB/2 ¹⁰	R/W
Only set the values listed above.					
b23 to b20	—	Reserved	These bits are 0 when read. Set them to 0 when writing.		R/W
b24	CLKEN	MMCIF Clock Output Control *1	0: MMCIF clock is not output from the MMC_CLK pin (low level output) 1: MMCIF clock is output from the MMC_CLK pin		R/W
b30 to b25	—	Reserved	These bits are 0 when read. Set them to 0 when writing.		R/W
b31	MMCBUSBSY	MMC Bus Status Flag	0: MMC bus is not busy 1: MMC bus is busy		R

Note 1. Do not rewrite the CLKEN or CLKDIV[3:0] bits when the MMCBUSBSY flag is 1.

The CECLKCTRL register controls the MMCIF clock and defines the timeout value. Do not rewrite the CECLKCTRL register during the command sequence.

MMCBUSBSY Flag (MMC Bus Status Flag)

When writing a value to the CECMDSET register and starting the command sequence, the CEHOSTSTS1.CMDSEQ flag and MMCBUSBSY flag become 1 simultaneously. When the command sequence is complete and the

CEHOSTSTS1.CMDSEQ flag becomes 0, 10 cycles of the MMCIF clock are output before the MMCBUSBSY flag becomes 0.

49.2.7 Buffer Access Setting Register (CEBUFACC)

Address(es): 0008 851Ch

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	DMATYP	DMAWEN	DMAREN	—	—	—	—	—	—	—	ATYP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

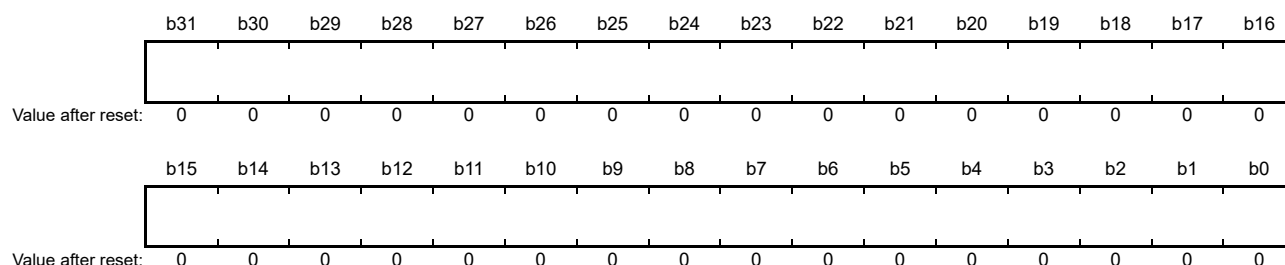
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b16	ATYP	Buffer Access Select	0: When accessing the CEDATA register, the byte endian is not switched 1: When accessing the CEDATA register, the byte endian is switched	R/W
b23 to b17	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b24	DMAREN	Enable DMA Transfer Request to Read the CEDATA Register	0: CEDATA register cannot be read using DMA transfer 1: CEDATA register can be read using DMA transfer	R/W
b25	DMAWEN	Enable DMA Transfer Request to Write to the CEDATA Register	0: CEDATA register cannot be written to using DMA transfer 1: CEDATA register can be written to using DMA transfer	R/W
b26	DMATYP	DMA Transfer Method Select	Set this bit to 1 when the DMAREN or DMAWEN bit is 1 (enabled).	R/W
b31 to b27	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W

The CEBUFACC register is used to select a method of accessing the CEDATA register, and to set the DMAC and DTC transfer method. Do not rewrite the CEBUFACC register during the command sequence. For details on the MMCIF buffer, refer to section 49.3.3, MMCIF Buffer Structure and Access Method.

49.2.8 Response Register 0 (CERESP0), Response Register 1 (CERESP1), Response Register 2 (CERESP2), Response Register 3 (CERESP3)

Address(es): CERESP3 0008 8520h, CERESP2 0008 8524h, CERESP1 0008 8528h, CERESP0 0008 852Ch



Registers CERESP0 to CERESP3 are read-only registers used for storing response values from the MMC. Depending on the type of response received, the MMCIF divides and stores the responses in their appropriate registers. Table 49.4 lists the Response Value Storage Destinations. Refer to section 49.3.1, Command and Response Formats for details on the format of the response value.

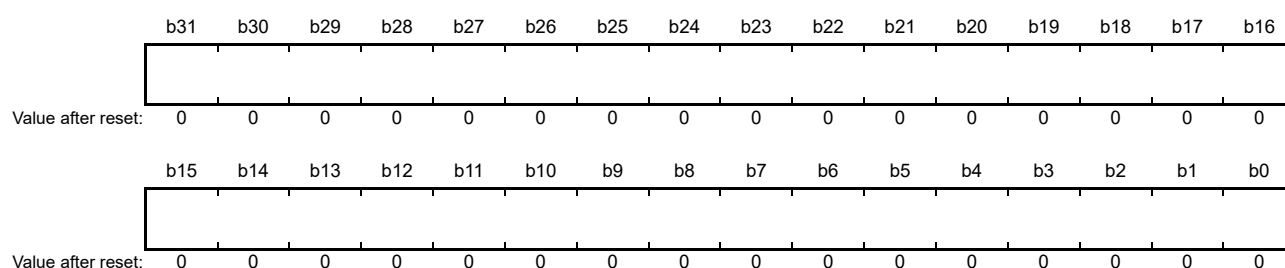
Table 49.4 Response Value Storage Destinations

Response Type	CERESP3 Register	CERESP2 Register	CERESP1 Register	CERESP0 Register
R1	—	—	—	[39:8]
R1b *1	—	—	—	[39:8]
R2	[127:96]	[95:64]	[63:32]	[31:0]
R3	—	—	—	[39:8]
R4	—	—	—	[39:8]
R5	—	—	—	[39:8]

Note 1. The response for an automatically issued CMD12 is stored in the CERESPCMD12 register.

49.2.9 Automatically Issued CMD12 Response Register (CERESPCMD12)

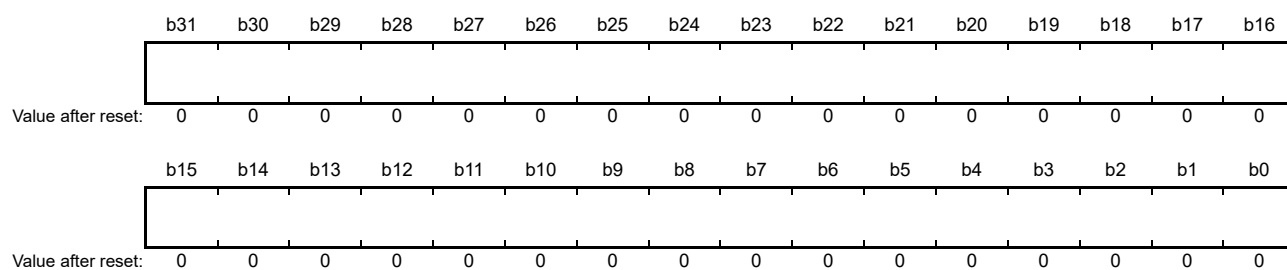
Address(es): 0008 8530h



CERESPCMD12 is a read-only register that stores the response from the MMC when CMD12 is automatically issued.

49.2.10 Data Register (CEDATA)

Address(es): 0008 8534h



The CEDATA data register is used for accessing the MMCIF buffer. Refer to section 49.3.2, Data Block Format for details on the format of the read and write data.

49.2.11 Boot Operation Setting Register (CEBOOT)

Address(es): 0008 853Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	SBTCLKDIV[3:0]				SBTACKTO[3:0]				SFSTBTDATTO[3:0]				SBTDATTO[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b19 to b16	SBTDATTO[3:0]	Setting for Timeout Between Boot Data	b19 b16 0 0 0 0: $2^{14} \times$ MMCIF clock cycle 0 0 0 1: $2^{15} \times$ MMCIF clock cycle 0 0 1 0: $2^{16} \times$ MMCIF clock cycle 0 0 1 1: $2^{17} \times$ MMCIF clock cycle 0 1 0 0: $2^{18} \times$ MMCIF clock cycle 0 1 0 1: $2^{19} \times$ MMCIF clock cycle 0 1 1 0: $2^{20} \times$ MMCIF clock cycle 0 1 1 1: $2^{21} \times$ MMCIF clock cycle b19 b16 1 0 0 0: $2^{22} \times$ MMCIF clock cycle 1 0 0 1: $2^{23} \times$ MMCIF clock cycle 1 0 1 0: $2^{24} \times$ MMCIF clock cycle 1 0 1 1: $2^{25} \times$ MMCIF clock cycle 1 1 0 0: $2^{26} \times$ MMCIF clock cycle 1 1 0 1: $2^{27} \times$ MMCIF clock cycle 1 1 1 0: $2^{28} \times$ MMCIF clock cycle 1 1 1 1: $2^{29} \times$ MMCIF clock cycle	R/W
b23 to b20	SFSTBTDATTO[3:0]	First Boot Data Timeout Setting	b23 b20 0 0 0 0: $2^{14} \times$ MMCIF clock cycle 0 0 0 1: $2^{15} \times$ MMCIF clock cycle 0 0 1 0: $2^{16} \times$ MMCIF clock cycle 0 0 1 1: $2^{17} \times$ MMCIF clock cycle 0 1 0 0: $2^{18} \times$ MMCIF clock cycle 0 1 0 1: $2^{19} \times$ MMCIF clock cycle 0 1 1 0: $2^{20} \times$ MMCIF clock cycle 0 1 1 1: $2^{21} \times$ MMCIF clock cycle b23 b20 1 0 0 0: $2^{22} \times$ MMCIF clock cycle 1 0 0 1: $2^{23} \times$ MMCIF clock cycle 1 0 1 0: $2^{24} \times$ MMCIF clock cycle 1 0 1 1: $2^{25} \times$ MMCIF clock cycle 1 1 0 0: $2^{26} \times$ MMCIF clock cycle 1 1 0 1: $2^{27} \times$ MMCIF clock cycle 1 1 1 0: $2^{28} \times$ MMCIF clock cycle 1 1 1 1: $2^{29} \times$ MMCIF clock cycle	R/W
b27 to b24	SBTACKTO[3:0]	Boot Acknowledge Timeout Setting	b27 b24 0 0 0 0: $2^{14} \times$ MMCIF clock cycle 0 0 0 1: $2^{15} \times$ MMCIF clock cycle 0 0 1 0: $2^{16} \times$ MMCIF clock cycle 0 0 1 1: $2^{17} \times$ MMCIF clock cycle 0 1 0 0: $2^{18} \times$ MMCIF clock cycle 0 1 0 1: $2^{19} \times$ MMCIF clock cycle 0 1 1 0: $2^{20} \times$ MMCIF clock cycle 0 1 1 1: $2^{21} \times$ MMCIF clock cycle b27 b24 1 0 0 0: $2^{22} \times$ MMCIF clock cycle 1 0 0 1: $2^{23} \times$ MMCIF clock cycle 1 0 1 0: $2^{24} \times$ MMCIF clock cycle 1 0 1 1: $2^{25} \times$ MMCIF clock cycle 1 1 0 0: $2^{26} \times$ MMCIF clock cycle 1 1 0 1: $2^{27} \times$ MMCIF clock cycle 1 1 1 0: $2^{28} \times$ MMCIF clock cycle 1 1 1 1: $2^{29} \times$ MMCIF clock cycle	R/W
b31 to b28	SBTCLKDIV[3:0]	MMCIF Clock Frequency During Boot Operations	b31 b28 0 0 0 0: PCLKB/2 0 0 0 1: PCLKB/4 0 0 1 0: PCLKB/8 0 0 1 1: PCLKB/16 Only set the values listed above.	R/W

The CEBOOT register is used for selecting the MMCIF clock frequency during boot operations, and for setting timeout values. Do not rewrite this register during a command sequence or during boot operations.

SBTCLKDIV[3:0] Bits (MMCIF Clock Frequency During Boot Operations)

The SBTCLKDIV[3:0] bit setting value must be the equal to or lower than the MMCIF clock frequency selected in the CECLKCTRL.CLKDIV[3:0] bits (the MMCIF clock frequency during boot operations must be equal to or greater than the MMCIF clock frequency in normal mode).

Refer to section 49.3.5, MMCIF Clock Frequency During Boot Operations for details on the MMCIF clock frequency during boot operation.

49.2.12 Interrupt Status Flag Register (CEINT)

Address(es): 0008 8540h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	CMD12 DRE	CMD12 RBE	CMD12 CRE	DTRAN E	BUFRE	BUFWE N	BUFRE N	—	—	RBSYE	CRSPE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
CMDVI O	BUFVI O	—	—	WDATE RR	RDATE RR	RIDX RR	RSPERR	—	—	—	CRCST O	WDATT O	RDATT O	RBSYT O	RSPTO
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	RSPTO	Response Timeout Flag	0: Response timeout did not occur 1: Response timeout occurred	R/(W) *1
b1	RBSYTO	Response Busy Timeout Flag	0: Response busy timeout did not occur 1: Response busy timeout occurred	R/(W) *1
b2	RDATTO	Read Data Timeout Flag	0: Read data timeout did not occur 1: Read data timeout occurred	R/(W) *1
b3	WDATTO	Write Data Timeout Flag	0: Write data timeout did not occur 1: Write data timeout occurred	R/(W) *1
b4	CRCSTO	CRC Status Timeout Flag	0: CRC status timeout did not occur 1: CRC status timeout occurred	R/(W) *1
b7 to b5	—	Reserved	These bits are 0 when read and cannot be modified.	R
b8	RSPERR	Response Error Flag	0: Response error did not occur 1: Response error occurred	R/(W) *1
b9	RIDXERR	Response Index Error Flag	0: Response index error did not occur 1: Response index error occurred	R/(W) *1
b10	RDATERR	Read Data Error Flag	0: Read data error did not occur 1: Read data error occurred	R/(W) *1
b11	WDATERR	Write Data Error Flag	0: Write data error did not occur 1: Write data error occurred	R/(W) *1
b13, b12	—	Reserved	These bits are 0 when read and cannot be modified.	R
b14	BUFVIO	MMCIF Buffer Access Error Flag	0: MMCIF buffer access error did not occur 1: MMCIF buffer access error occurred	R/(W) *1
b15	CMDVIO	Command Issue Error Flag	0: Command issuance error did not occur 1: Command issuance error occurred	R/(W) *1
b16	CRSPE	Command Response Complete Flag	0: Command transmission or response reception incomplete 1: Command transmission or response reception complete	R/(W) *1
b17	RBSYE	Response Busy Complete Flag	0: Response busy reception incomplete 1: Response busy reception complete	R/(W) *1
b19, b18	—	Reserved	These bits are 0 when read and cannot be modified.	R
b20	BUFREN	MMCIF Buffer Read Enable Flag	0: Reading data from MMCIF buffer is disabled 1: Reading data from MMCIF buffer is enabled	R/(W) *1
b21	BUFWE	MMCIF Buffer Write Enable Flag	0: Writing data to MMCIF buffer is disabled 1: Writing data to MMCIF buffer is enabled	R/(W) *1
b22	BUFRE	MMCIF Buffer Read Complete Flag	0: Reading data from MMCIF buffer is incomplete 1: Reading data from MMCIF buffer is complete	R/(W) *1
b23	DTRAN	Data Transmit Complete Flag	0: Data transmission incomplete 1: Data transmission complete	R/(W) *1
b24	CMD12CRE	Automatically Issued CMD12 Response Complete Flag	0: CMD12 automatic issuance incomplete 1: CMD12 automatic issuance complete	R/(W) *1
b25	CMD12RBE	Automatically Issued CMD12 Response Busy Complete Flag	0: Response reception and response busy triggered by automatically issued CMD12 are incomplete 1: Response reception and response busy triggered by automatically issued CMD12 are complete	R/(W) *1

Bit	Symbol	Bit Name	Description	R/W
b26	CMD12DRE	Automatically Issued CMD12 Response Busy and MMCIF Buffer Read Complete Flag	0: Response busy and MMCIF buffer read access triggered by automatically issued CMD12 are incomplete 1: Response busy and MMCIF buffer read access triggered by automatically issued CMD12 are complete	R/(W) *1
b31 to b27	—	Reserved	These bits are 0 when read and cannot be modified.	R

Note 1. This flag cannot be set to 1 by the user. This flag becomes 0 when set to 0 by the user.

The CEINT register indicates the various states in the command sequence. When clearing a flag, set the flag to be cleared to 0 and set the other flags to 1. Refer to section 49.3.8, MMCIF Processing When an Error or Timeout Occurs for details on MMCIF operation when errors and timeouts occur.

RSPTO Flag (Response Timeout Flag)

This flag indicates that a response or boot acknowledge was not received within the specified period. Note that the command sequence does not stop even if the RSPTO flag becomes 1.

- When not performing a boot operation, this flag becomes 1 under the following condition:
 - After a command is issued (including an automatically issued CMD12), a response is not received even after the number of clocks selected in the CECLKCTRL.SRSP TO[1:0] bits have been output.
- When performing a boot operation, this flag becomes 1 under the following condition:
 - When the CECMDSET.BOOTACK bit is set to 1 (boot acknowledge received), a boot acknowledge is not received even after the number of clocks selected in the CEBOOT.SBTACKTO[3:0] bits have been output.
- This flag becomes 0 under the following condition:
 - The flag is set to 0.

RBSYTO Flag (Response Busy Timeout Flag)

This flag indicates that the response is in a busy state longer than the specified period. Note that the command sequence does not stop even if the RBSYTO flag becomes 1.

- This flag becomes 1 under the following condition:
 - After a command is issued (including an automatically issued CMD12), the response is in the busy state even after the number of clocks selected in the CECLKCTRL.SRBSYTO[3:0] bits have been output.
- This flag becomes 0 under the following condition:
 - The flag is set to 0.

RDATTO Flag (Read Data Timeout Flag)

This flag indicates that data was not received within the specified period. Note that the command sequence does not stop even if the RDATTO flag becomes 1.

- When not performing a boot operation, this flag becomes 1 under any of the following conditions:
 - After a read command is issued, data is not received even after the number of clocks selected in the CECLKCTRL.SRWDTO[3:0] bits have been output.
 - After read data is received, data is not received even after the number of clocks selected in the CECLKCTRL.SRWDTO[3:0] bits have been output.
- When performing a boot operation, this flag becomes 1 under any of the following conditions:
 - The first data is not received even after the number of clocks selected in the CEBOOT.SFSTBT DATTO[3:0] bits have been output.
 - After read data is received, the next data is not received even after the number of clocks selected in the CEBOOT.SBT DATTO[3:0] bits have been output.
- This flag becomes 0 under the following condition:
 - The flag is set to 0.

WDATTO Flag (Write Data Timeout Flag)

During a data write access, this flag indicates that the MMC was in the busy state longer than the specified period. Note that the command sequence does not stop even when the WDATTO flag becomes 1.

— This flag becomes 1 under the following condition:

- After a CRC status token is received, the MMC is in a busy state even after the number of clocks selected in the CCLKCTRL.SRWDTO[3:0] bits have been output.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

CRCSTO Flag (CRC Status Timeout Flag)

This flag indicates that a CRC status token was not received. Note that the command sequence does not stop even if the CRCSTO flag becomes 1.

— This flag becomes 1 under the following condition:

- A CRC status token is not received.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

RSPERR Flag (Response Error Flag)

This flag indicates that the response or boot acknowledge is in error. Note that the command sequence stops when the RSPERR flag becomes 1.

— When not performing a boot operation, this flag becomes 1 under any of the following conditions:

- The response's transmission bit is 1.
- The response's end bit is in error.
- Bits [7:1] of the response are in error (content to be checked is specified in the CECMDSET.RCRC7C[1:0] bits).

— When performing a boot operation, this flag becomes 1 under any of the following conditions:

- The bit pattern of the boot acknowledge is in error.
- The end bit of the boot acknowledge is in error.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

RIDXERR Flag (Response Index Error Flag)

This flag indicates that the command index or check bits for the response is in error. Note that the command sequence stops when the RIDXERR flag becomes 1.

— This flag becomes 1 under the following condition:

- The response's command index field value or check bits field value is in error (content to be checked is specified in the CECMDSET.RIDXC[1:0] bits).

— This flag becomes 0 under the following condition:

- The flag is set to 0.

RDATERR Flag (Read Data Error Flag)

This flag indicates that the read data is in error. Note that the command sequence stops when the RDATERR flag becomes 1.

— This flag becomes 1 under any of the following conditions:

- CMD16 of the read data is in error.
- The end bit of the read data is in error.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

WDATERR Flag (Write Data Error Flag)

This flag indicates that the write data is in error. Note that the command sequence stops when the WDATERR flag becomes 1.

— This flag becomes 1 under any of the following conditions:

- The status of the CRC status token is in error.
- The end bit of the CRC status token is in error.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

BUFVIO Flag (MMCIF Buffer Access Error Flag)

This flag indicates illegal accesses to the MMCIF buffer. Note that the command sequence does not stop even if the BUFVIO flag becomes 1.

— This flag becomes 1 under any of the following conditions:

- The CEDATA register was accessed more times than the block size set in the CEBLOCKSET.BLKSIZ[15:0] bits.
- Despite the BUFREN bit not being set to 1 (during DMA transfer, the MMCIF buffer read DMA transfer is not requested), the CEDATA register was accessed while data was being read from the MMC.
- Despite the BUFWEN bit not being set to 1 (during DMA transfer, the MMCIF buffer write DMA transfer is not requested), the CEDATA register was accessed while data was being written to the MMC.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

Note: If the MMCIF buffer is accessed after a communication error occurs, the BUFVIO flag may become 1.

CMDVIO Flag (Command Issue Error Flag)

This flag indicates that registers CECMDSET and CEBLOCKSET were set illegally.

— This flag becomes 1 under any of the following conditions:

- A value is set to the CECMDSET.CMD[5:0] bits during a command sequence (the command sequence does not automatically stop).
- A value is set to the CECMDSET.CMD[5:0] bits when one of the following is in agreement with the CECMDSET and CEBLOCKSET register setting:
 - The RTYP[1:0] bits are 00b (no response), and the RBSY bit is 1 (response busy).
 - The RTYP[1:0] bits are 00b, the WDAT bit is 1 (data present), and the BOOT bit is 0 (command sequence when not performing boot operations).
 - The WDAT bit is 0 (no data present), and the CMD12EN bit is 1 (CMD12 is automatically issued).
 - The WDAT bit is 1, the CMLTE bit is 0 (single block transfer), and the CMD12EN bit is 1.
 - The WDAT bit is 1, the RBSY bit is 1 (response busy), and the CMD12EN bit is 1.
 - The WDAT bit is 1, and the BLKSIZ[15:0] bits (transfer block size) are 00h.
 - The WDAT bit is 1, and the BLKSIZ[15:0] bits are greater than 512.
 - The WDAT bit is 1, the CMLTE bit is 1 (multiple block transfer), and the BLKCNT[15:0] bits (number of transfer blocks) are 00h.
 - The BOOT bit is 1 (command sequence when performing boot operations), and the WDAT bit is 0 (no data present).
 - The BOOT bit is 1, and the DWEN bit is 1 (write data to the MMC).
 - The BOOT bit is 1, and the RBSY bit is 1.
 - The BOOT bit is 1, and the CMD12EN bit is 1.
 - The BOOTACK bit is 1 (boot acknowledge received), and the BOOT bit is 0 (command sequence when not performing boot operations).

— This flag becomes 0 under the following condition:

- The flag is set to 0.

CRSPE Flag (Command Response Complete Flag)

This flag indicates when command issuance or response reception is complete. The response complete flag for the automatically issued CMD12 is the CMD12CRE flag.

- When not performing a boot operation, this flag becomes 1 under any of the following conditions:
 - A command is issued when no response is configured.
 - A response is received when a 6-byte response and 17-byte response are set.
- When performing a boot operation, this flag becomes 1 under the following condition:
 - A boot acknowledge is received while the CECMDSET.BOOTACK bit is set to 1 (boot acknowledge received).
- This flag becomes 0 under the following condition:
 - The flag is set to 0.

RBSYE Flag (Response Busy Complete Flag)

This flag indicates that the response busy is complete. Note that when the RBSYE flag becomes 1, the CRSPE flag also becomes 1. The response busy complete flag for the automatically issued CMD12 is the CMD12RBE flag.

- This flag becomes 1 under the following condition:
 - Response reception and response busy reception are complete.
- This flag becomes 0 under the following condition:
 - The flag is set to 0.

BUFREN Flag (MMCIF Buffer Read Enable Flag)

This flag indicates that the MMCIF buffer has become read accessible. When using the CPU to read data from the CEDATA register, set the BUFREN flag to 0 before reading the amount of data set in the CEBLOCKSET.BLKSIZ[15:0] bits. If the CEBUFACC.DMAREN bit is 1, the BUFREN flag will not become 1.

- This flag becomes 1 under the following condition:
 - The transfer block size amount of data is stored in the MMCIF buffer and the MMCIF buffer becomes read accessible.
- This flag becomes 0 under the following condition:
 - The flag is set to 0.

BUFWEN Flag (MMCIF Buffer Write Enable Flag)

This flag indicates that the MMCIF buffer has become write accessible. When using the CPU to write data to the CEDATA register, set the BUFWEN flag to 0 before writing the amount of data set in the CEBLOCKSET.BLKSIZ[15:0] bits. If the CEBUFACC.DMAWEN bit is 1, the BUFWEN flag will not become 1.

- This flag becomes 1 under the following condition:
 - The MMCIF buffer becomes empty and becomes write accessible.
- This flag becomes 0 under the following condition:
 - The flag is set to 0.

BUFRE Flag (MMCIF Buffer Read Complete Flag)

This flag indicates that the read access of the MMCIF is complete.

- When not performing a boot operation, this flag becomes 1 under the following condition:
 - All data in all blocks is received, the read access of the MMCIF buffer is finished, and the MMCIF buffer is empty.
- When performing a boot operation, this flag becomes 1 under the following condition:
 - All data in all blocks is received, the read access of the MMCIF buffer is finished, the MMC_CMD pin changes from low to high, and the amount of time it takes 48 MMCIF clock cycles to elapse has passed.
- This flag becomes 0 under the following condition:
 - The flag is set to 0.

DTRANE Flag (Data Transmit Complete Flag)

This flag indicates that all data in all blocks has been transmitted.

— This flag becomes 1 under any of the following conditions:

- When receiving a CRC status token, the busy status is complete after the CRC status token is received.
- When not receiving a CRC status token, data transmission is complete.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

CMD12CRE Flag (Automatically Issued CMD12 Response Complete Flag)

This flag indicates that the response for the automatically issued CMD12 was received.

— This flag becomes 1 under the following condition:

- A response is received for the automatically issued CMD12.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

CMD12RBE Flag (Automatically Issued CMD12 Response Busy Complete Flag)

This flag indicates that the response busy for the automatically issued CMD12 is complete. Note that when the CMD12RBE flag becomes 1, the CMD12CRE flag also becomes 1. When the CMD12RBE flag becomes 1 during a multiple block write, the DTRANE flag also becomes 1.

— This flag becomes 1 under the following condition:

- A response is received for the automatically issued CMD12, and the response busy is complete.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

CMD12DRE Flag (Automatically Issued CMD12 Response Busy and MMCIF Buffer Read Complete Flag)

This flag indicates that the response busy for the automatically issued CMD12 is complete, and the MMCIF buffer read access is complete. Note that when the CMD12DRE flag becomes 1, the CMD12RBE, CMD12CRE, and BUFRE flags also become 1.

— This flag becomes 1 under the following condition:

- The response busy for the automatically issued CMD12 is complete, and the read access from the MMCIF buffer is complete.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

49.2.13 Interrupt Request Enable Register (CEINTEN)

Address(es): 0008 8544h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	—	—	MCMD12DRE	MCMD12RBE	MCMD12CRE	MDTRAN	MBUFRE	MBUFWEN	MBUFREN	—	—	MRBSYE	MCRSPE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
MCMDVIO	MBUFVIO	—	—	MWDATERR	MRDATERR	MRIDXERR	MRSPEERR	—	—	—	MCRCSTO	MWDATTO	MRDATTO	MRBSYTO	MRSPTO
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

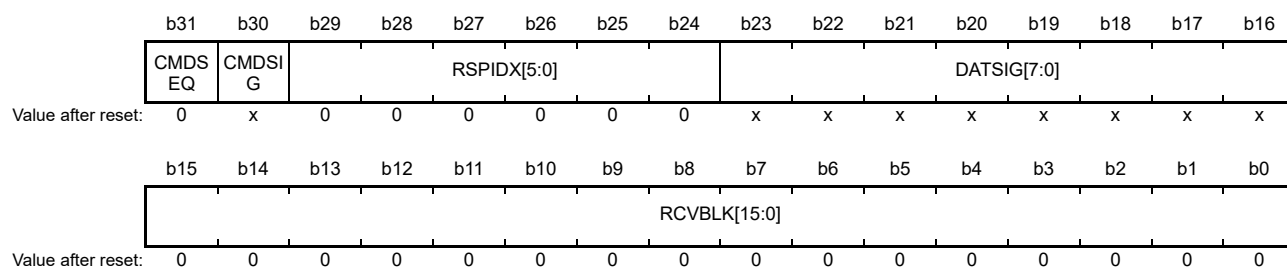
Bit	Symbol	Bit Name	Description	R/W
b0	MRSPTO	RSPTO Interrupt Request Enable	0: RSPTO flag interrupt requests disabled 1: RSPTO flag interrupt requests enabled	R/W
b1	MRBSYTO	RBSYTO Interrupt Request Enable	0: RBSYTO flag interrupt requests disabled 1: RBSYTO flag interrupt requests enabled	R/W
b2	MRDATTO	RDATTO Interrupt Request Enable	0: RDATTO flag interrupt requests disabled 1: RDATTO flag interrupt requests enabled	R/W
b3	MWDATTO	WDATTO Interrupt Request Enable	0: WDATTO flag interrupt requests disabled 1: WDATTO flag interrupt requests enabled	R/W
b4	MCRCSTO	CRCSTO Interrupt Request Enable	0: CRCSTO flag interrupt requests disabled 1: CRCSTO flag interrupt requests enabled	R/W
b7 to b5	—	Reserved	These bits are 0 when read and cannot be modified.	R
b8	MRSPEERR	RSPERR Interrupt Request Enable	0: RSPERR flag interrupt requests disabled 1: RSPERR flag interrupt requests enabled	R/W
b9	MRIDXERR	RIDXERR Interrupt Request Enable	0: RIDXERR flag interrupt requests disabled 1: RIDXERR flag interrupt requests enabled	R/W
b10	MRDATERR	RDATERR Interrupt Request Enable	0: RDATERR flag interrupt requests disabled 1: RDATERR flag interrupt requests enabled	R/W
b11	MWDATERR	WDATERR Interrupt Request Enable	0: WDATERR flag interrupt requests disabled 1: WDATERR flag interrupt requests enabled	R/W
b13, b12	—	Reserved	These bits are 0 when read and cannot be modified.	R
b14	MBUFVIO	BUFVIO Interrupt Request Enable	0: BUFVIO flag interrupt requests disabled 1: BUFVIO flag interrupt requests enabled	R/W
b15	MCMDVIO	CMDVIO Interrupt Request Enable	0: CMDVIO flag interrupt requests disabled 1: CMDVIO flag interrupt requests enabled	R/W
b16	MCRSPE	CRSPE Interrupt Request Enable	0: CRSPE flag interrupt requests disabled 1: CRSPE flag interrupt requests enabled	R/W
b17	MRBSYE	RBSYE Interrupt Request Enable	0: RBSYE flag interrupt requests disabled 1: RBSYE flag interrupt requests enabled	R/W
b19, b18	—	Reserved	These bits are 0 when read and cannot be modified.	R
b20	MBUFREN	BUFREN Interrupt Request Enable	0: BUFREN flag interrupt requests disabled 1: BUFREN flag interrupt requests enabled	R/W
b21	MBUFWEN	BUFWEN Interrupt Request Enable	0: BUFWEN flag interrupt requests disabled 1: BUFWEN flag interrupt requests enabled	R/W
b22	MBUFRE	BUFRE Interrupt Request Enable	0: BUFRE flag interrupt requests disabled 1: BUFRE flag interrupt requests enabled	R/W
b23	MDTRAN	DTRAN Interrupt Request Enable	0: DTRAN flag interrupt requests disabled 1: DTRAN flag interrupt requests enabled	R/W
b24	MCMD12CRE	CMD12CRE Interrupt Request Enable	0: CMD12CRE flag interrupt requests disabled 1: CMD12CRE flag interrupt requests enabled	R/W
b25	MCMD12RBE	CMD12RBE Interrupt Request Enable	0: CMD12RBE flag interrupt requests disabled 1: CMD12RBE flag interrupt requests enabled	R/W
b26	MCMD12DRE	CMD12DRE Interrupt Request Enable	0: CMD12DRE flag interrupt requests disabled 1: CMD12DRE flag interrupt requests enabled	R/W
b31 to b27	—	Reserved	These bits are 0 when read and cannot be modified.	R

When the status flags in the CEINT register become 1, the CEINTEN register controls whether or not to generate an

interrupt request. When bits in this register are 1 and the corresponding status flag in the CEINT register is 1, then an interrupt request is generated. Refer to section 49.5, Interrupts for details on interrupt requests.

49.2.14 Status Register 1 (CEHOSTSTS1)

Address(es): 0008 8548h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	RCVBLK[15:0]	Number of Blocks Transferred Flag	Indicates the number of blocks transferred	R
b23 to b16	DATSIG[7:0]	MMC_D[7:0] Pin Status Flag	Indicates the status of pins MMC_D7 to MMC_D0 0: Pin level is low 1: Pin level is high	R
b29 to b24	RSPIDX[5:0]	Response Index	Indicates the command index field value or the check bits field value of the response	R
b30	CMDSIG	MMC_CMD Pin Status Flag	Indicates the status of the MMC_CMD pin 0: Pin level is low 1: Pin level is high	R
b31	CMDSEQ	Command Sequence Status Flag	0: Command sequence is in its initial state 1: Command sequence in progress	R

The CEHOSTSTS1 register indicates the number of blocks that have been transferred, indicates the status of the MMC_CMD pin and pins MMC_D0 to MMC_D7, indicates the index of the response received, and indicates the status of the command sequence.

RCVBLK[15:0] Bits (Number of Blocks Transferred Flag)

These bits indicate the number of blocks that have been transferred.

When the CECMDSET.DWEN bit is 0, these bits indicate the number of blocks read from the MMC.

When the CECMDSET.DWEN bit is 1, these bits indicate the number of blocks written to the MMC.

DATSIG[7:0] Bits (MMC_D[7:0] Pin Status Flag)

These bits indicate the status of pins MMC_D0 to MMC_D7.

Note: If a communication error or timeout occurs, the MMC_D0 pin may remain low.

49.2.15 Status Register 2 (CEHOSTSTS2)

Address(es): 0008 854Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	CRCST E	CRC16 E	AC12C RCE	RSPCR C7E	CRCST EBE	RDATE BE	AC12R EBE	RSPEB E	AC12ID XE	RSPID XE	BTACK PATE	BTACK EBE	—	CRCST[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	STRDA TTO	DATBS YTO	CRCST TO	AC12B SYTO	RSPBS YTO	AC12R SPTO	STRSP TO	BTACK TO	FSTBT DATTO	BTDAT TO	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b5	BTDATTO	Boot Operation Timeout Flag	0: Boot operation timeout did not occur 1: Boot operation timeout occurred	R
b6	FSTBTDATTO	First Boot Data Timeout Flag	0: First boot data timeout did not occur 1: First boot data timeout occurred	R
b7	BTACKTO	Boot Acknowledge Timeout Flag	0: Boot acknowledge timeout did not occur 1: Boot acknowledge timeout occurred	R
b8	STRSPTO	Response Timeout Flag	0: Response timeout did not occur 1: Response timeout occurred	R
b9	AC12RSPTO	Automatically Issued CMD12 Response Timeout Flag	0: Automatically issued CMD12 response timeout did not occur 1: Automatically issued CMD12 response timeout occurred	R
b10	RSPBSYTO	Response Busy Timeout Flag	0: Response busy timeout did not occur 1: Response busy timeout occurred	R
b11	AC12BSYTO	Automatically Issued CMD12 Response Busy Timeout Flag	0: Automatically issued CMD12 response busy timeout did not occur 1: Automatically issued CMD12 response busy timeout occurred	R
b12	CRCSTTO	CRC Status Timeout Flag	0: CRC status timeout did not occur 1: CRC status timeout occurred	R
b13	DATBSYTO	Data Busy Timeout Flag	0: Data busy timeout did not occur 1: Data busy timeout occurred	R
b14	STRDATTO	Read Data Timeout Flag	0: Read data timeout did not occur 1: Read data timeout occurred	R
b15	—	Reserved	This bit is 0 when read. Set it to 0 when writing.	R/W
b18 to b16	CRCST[2:0]	CRC Status Indicator Flag	Indicates the CRC status	R
b19	—	Reserved	This bit is 0 when read. Set it to 0 when writing.	R/W
b20	BTACKEBE	Boot Acknowledge End Bit Error Flag	0: Boot acknowledge end bit error did not occur 1: Boot acknowledge end bit error occurred	R
b21	BTACKPATE	Boot Acknowledge Pattern Error Flag	0: Boot acknowledge patter error did not occur 1: Boot acknowledge patter error occurred	R
b22	RSPIDX	Command Response Index Error Flag (Other Than Automatically Issued CMD12)	0: Command response index error did not occur 1: Command response index error occurred	R
b23	AC12IDX	Automatically Issued CMD12 Response Index Error Flag	0: Automatically issued CMD12 response index error did not occur 1: Automatically issued CMD12 response index error occurred	R
b24	RSPEBE	Command Response End Bit Error Flag (Other Than Automatically Issued CMD12)	0: Command response end bit error did not occur 1: Command response end bit error occurred	R
b25	AC12REBE	Automatically Issued CMD12 Response End Bit Error Flag	0: Automatically issued CMD12 response end bit error did not occur 1: Automatically issued CMD12 response end bit error occurred	R
b26	RDATEBE	Read Data End Bit Error Flag	0: Read data end bit error did not occur 1: Read data end bit error occurred	R
b27	CRCSTEBE	CRC Status End Bit Error Flag	0: CRC status end bit error did not occur 1: CRC status end bit error occurred	R

Bit	Symbol	Bit Name	Description	R/W
b28	RSPCRC7E	Command Response CMD7 Error Flag (Other Than Automatically Issued CMD12)	0: Command response CMD7 error did not occur 1: Command response CMD7 error occurred	R
b29	AC12CRCE	Automatically Issued CMD12 Response CMD7 Error Flag	0: Automatically issued CMD12 response CMD7 error did not occur 1: Automatically issued CMD12 response CMD7 error occurred	R
b30	CRC16E	Read Data CMD16 Error Flag	0: Read data CMD16 error did not occur 1: Read data CMD16 error occurred	R
b31	CRCSTE	CRC Status Error Flag	0: CRC status error did not occur 1: CRC status error occurred	R

The CEHOSTSTS2 register indicates the status of various timeouts and errors.

BTDATTO Flag (Boot Operation Timeout Flag)

When performing boot operations, after read data is received, if the next data is not received even after the period set in the CEBOOT.SBTDATTO[3:0] bits elapses, this flag becomes 1.

FSTBTDATTO Flag (First Boot Data Timeout Flag)

When performing boot operations, if the first data is not received even after the period set in the CEBOOT.SFSTBTDATTO[3:0] bits elapses, this flag becomes 1.

BTACKTO Flag (Boot Acknowledge Timeout Flag)

When performing boot operations, if the boot acknowledge is not received even after the period set in the CEBOOT.SBTDATTO[3:0] bits elapses, this flag becomes 1.

STRSPTO Flag (Response Timeout Flag)

After issuing a command other than the automatically issued CMD12, if the response is not received even after the period set in the CECLKCTRL.SRSPTO[1:0] bits elapses, this flag becomes 1.

AC12RSPTO Flag (Automatically Issued CMD12 Response Timeout Flag)

After CMD12 is automatically issued, if the response is not received even after the period set in the CECLKCTRL.SRSPTO[1:0] bits elapses, this flag becomes 1.

RSPBSYTO Flag (Response Busy Timeout Flag)

After issuing a command other than the automatically issued CMD12, if the busy status continues even after the period set in the CECLKCTRL.SRBSYTO[3:0] bits elapses, this flag becomes 1.

AC12BSYTO Flag (Automatically Issued CMD12 Response Busy Timeout Flag)

After CMD12 is automatically issued, if the busy status continues even after the period set in the CECLKCTRL.SRBSYTO[3:0] bits elapses, this flag becomes 1.

CRCSTTO Flag (CRC Status Timeout Flag)

This flag becomes 1 if the CRC status token is not received.

DATBSYTO Flag (Data Busy Timeout Flag)

After the CRC status token is received, if the busy status continues even after the period set in the CECLKCTRL.SRWDTO[3:0] bits elapses, this flag becomes 1.

STRDATTO Flag (Read Data Timeout Flag)

After a read command is issued, if data is not received even after the period set in the CECLKCTRL.SRWDTO[3:0] bits elapses, this flag becomes 1. After data is received, if the next data is not received even after the period set in the CECLKCTRL.SRWDTO[3:0] bits elapses, this flag becomes 1.

CRCST[2:0] Flags (CRC Status Indicator Flag)

When the CECMDSET.BOOT bit is 0, these bits indicate the status of the received CRC status token.

When the CECMDSET.BOOT bit is 1, these bits indicate the bit pattern of the boot acknowledge.

BTACKEBE Flag (Boot Acknowledge End Bit Error Flag)

This flag becomes 1 if the end bit of the boot acknowledge is in error.

BTACKPATE Flag (Boot Acknowledge Pattern Error Flag)

This flag becomes 1 if the bit pattern of the boot acknowledge is in error.

RSPIDX Flag (Command Response Index Error Flag (Other Than Automatically Issued CMD12))

If the command index field value of the response, or the check bits field value is in error, this flag becomes 1. The content of the error is defined in the CECMDSET.RIDXC[1:0] bits.

AC12IDX Flag (Automatically Issued CMD12 Response Index Error Flag)

If the command index field value of the response for the automatically issued CMD12 is in error, this flag becomes 1.

The content of the error is defined in the CECMDSET.RIDXC[1:0] bits.

RSPEBE Flag (Command Response End Bit Error Flag (Other Than Automatically Issued CMD12))

If the end bit of the response for a command other than the automatically issued CMD12 is in error, this flag becomes 1.

AC12REBE Flag (Automatically Issued CMD12 Response End Bit Error Flag)

This flag becomes 1 if the end bit of the response for the automatically issued CMD12 is in error.

RDATEBE Flag (Read Data End Bit Error Flag)

This flag becomes 1 if the end bit for the read data is in error.

CRCSTEBE Flag (CRC Status End Bit Error Flag)

This flag becomes 1 if the end bit of the CRC status token is in error.

RSPCRC7E Flag (Command Response CMD7 Error Flag (Other Than Automatically Issued CMD12))

If bits [7:1] of a response for a command other than the automatically issued CMD12 are in error, this flag becomes 1.

The content of the error is stored in the CECMDSET.RCRC7C[1:0] bits.

AC12CRCE Flag (Automatically Issued CMD12 Response CMD7 Error Flag)

If bits [7:1] of the response for the automatically issued CMD12 are in error, this bit becomes 1. The content of the error

is stored in the CECMDSET.RCRC7C[1:0] bits.

CRC16E Flag (Read Data CMD16 Error Flag)

This flag becomes 1 when CRC16 of the read data is in error.

CRCSTE Flag (CRC Status Error Flag)

This flag becomes 1 when the CRC status value is in error.

49.2.16 MMC Detection and Port Control Register (CEDETECT)

Address(es): 0008 8570h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	CDSIG	CDRISE	CDFALL	—	—	—	—	—	—	MCDRISE	MCDFALL	—	—	—	—
Value after reset:	0	x	0	0	0	x	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b4	MCDFALL	CDFALL Interrupt Request Enable	0: CDFALL flag interrupt request disabled 1: CDFALL flag interrupt request enabled	R/W
b5	MCDRISE	CDRISE Interrupt Request Enable	0: CDRISE flag interrupt request disabled 1: CDRISE flag interrupt request enabled	R/W
b9 to b6	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b10	—	Reserved	This bit is undefined when read. Set it to 0 when writing.	R/W
b11	—	Reserved	This bit is 0 when read. Set it to 0 when writing.	R/W
b12	CDFALL	MMC_CD Pin Falling Edge Detection Flag	0: Falling edge on the MMC_CD pin not detected 1: Falling edge on the MMC_CD pin detected	R/(W) *1
b13	CDRISE	MMC_CD Pin Rising Edge Detection Flag	0: Rising edge on the MMC_CD pin not detected 1: Rising edge on the MMC_CD pin detected	R/(W) *1
b14	CDSIG	MMC_CD Pin Status Indication Flag	This flag indicates the status of the MMC_CD pin. 0: Pin level is low 1: Pin level is high	R
b31 to b15	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W

Note 1. This flag cannot be set to 1 by the user. This flag becomes 0 when set to 0 by the user.

The CEDETECT register controls detection of the MMC. Refer to section 49.5, Interrupts for details on the MMC detection interrupt request.

CDFALL Flag (MMC_CD Pin Falling Edge Detection Flag)

— This flag becomes 1 under the following condition:

- The MMC_CD pin changes from high to low.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

CDRISE Flag (MMC_CD Pin Rising Edge Detection Flag)

— This flag becomes 1 under the following condition:

- The MMC_CD pin changes from low to high.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

49.2.17 Special Mode Setting Register (CEADDMODE)

Address(es): 0008 8574h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	RESNO UT	—	CLKMA IN	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b18 to b0	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b19	CLKMAIN	MMCIF Internal Clock Control	0: Normal mode 1: Low power consumption mode (possible only when MMC is detected)	R/W
b20	—	Reserved	This bit is 0 when read. Set it to 0 when writing.	R/W
b21	RESNOUT	Reset Output *1	0: MMC_RES# pin is high 1: MMC_RES# pin is low	R/W
b31 to b22	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W

Note 1. The same functions can be performed using a general-use I/O port.

The CEADDMODE register controls the MMCIF internal clock.

49.2.18 Version Register (CEVERSION)

Address(es): 0008 857Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	SWRS T	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	VERSION[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	VERSION[15:0]	Version Indicator	These bits indicate the version of the MMCIF.	R
b30 to b16	—	Reserved	These bits are 0 when read. Set them to 0 when writing.	R/W
b31	SWRST	MMCIF Software Reset	0: MMCIF exits the MMCIF software reset and enters normal operation 1: MMCIF is in the MMCIF software reset state	R/W

The CEVERSION register indicates the MMCIF version and controls the MMCIF software reset.

SWRST Bit (MMCIF Software Reset)

When the SWRST bit is set to 1, the MMCIF is reset, and all register values in the MMCIF become their value after reset (except for the SWRST bit).

49.3 MMCIF Operation

49.3.1 Command and Response Formats

Figure 49.2 shows the Command Format. The command index field value set in the CECMDSET.CMD[5:0] bits is reflected in bits [45:40] of the command, and the argument field value set in the CEARG register is reflected in bits [39:8] of the command.

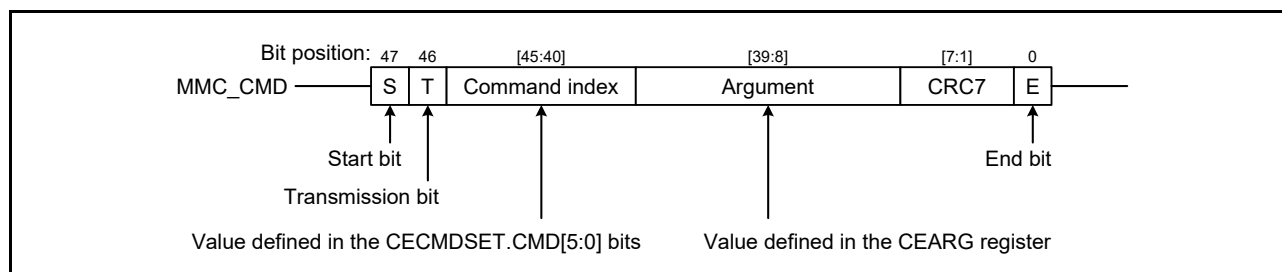


Figure 49.2 Command Format

Figure 49.3 shows the format of a 6-byte response (response other than the R2 response), and Figure 49.4 shows the format of a 17-byte response (R2 response). The command index field value or check bits field value of the response is stored in the CEHOSTSTS1.RSPIDX[5:0] flags. For a 6-byte response, bits [39:8] of the response are stored in the CERESP0 register, and for a 17-byte response, bits [127:0] of the response are stored in registers CERESP3 to CERESP0.

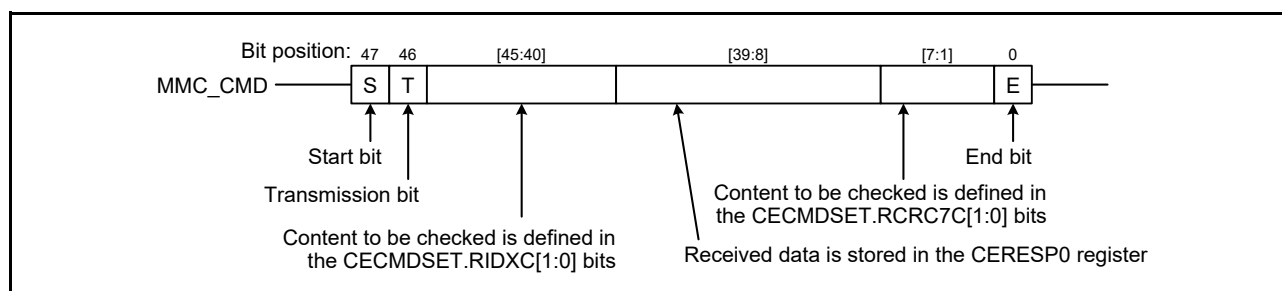


Figure 49.3 6-Byte Response Format (Response Other Than the R2 Response)

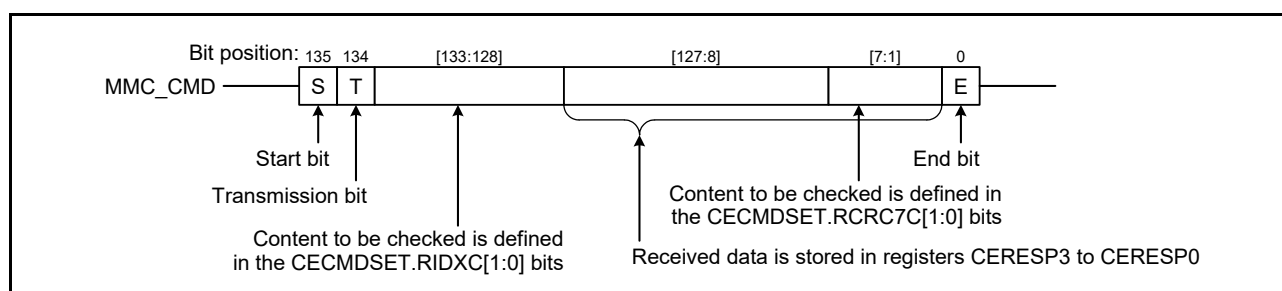


Figure 49.4 17-Byte Response Format (R2 Response)

49.3.2 Data Block Format

Figure 49.5 shows the data block format. Refer to section 49.3.3, MMCIF Buffer Structure and Access Method for details on data 0 to data 3 shown in the figure. When writing data to an MMC, data stored in the MMCIF buffer is transmitted. When reading data from an MMC, data received from the MMC is stored in the MMCIF buffer.

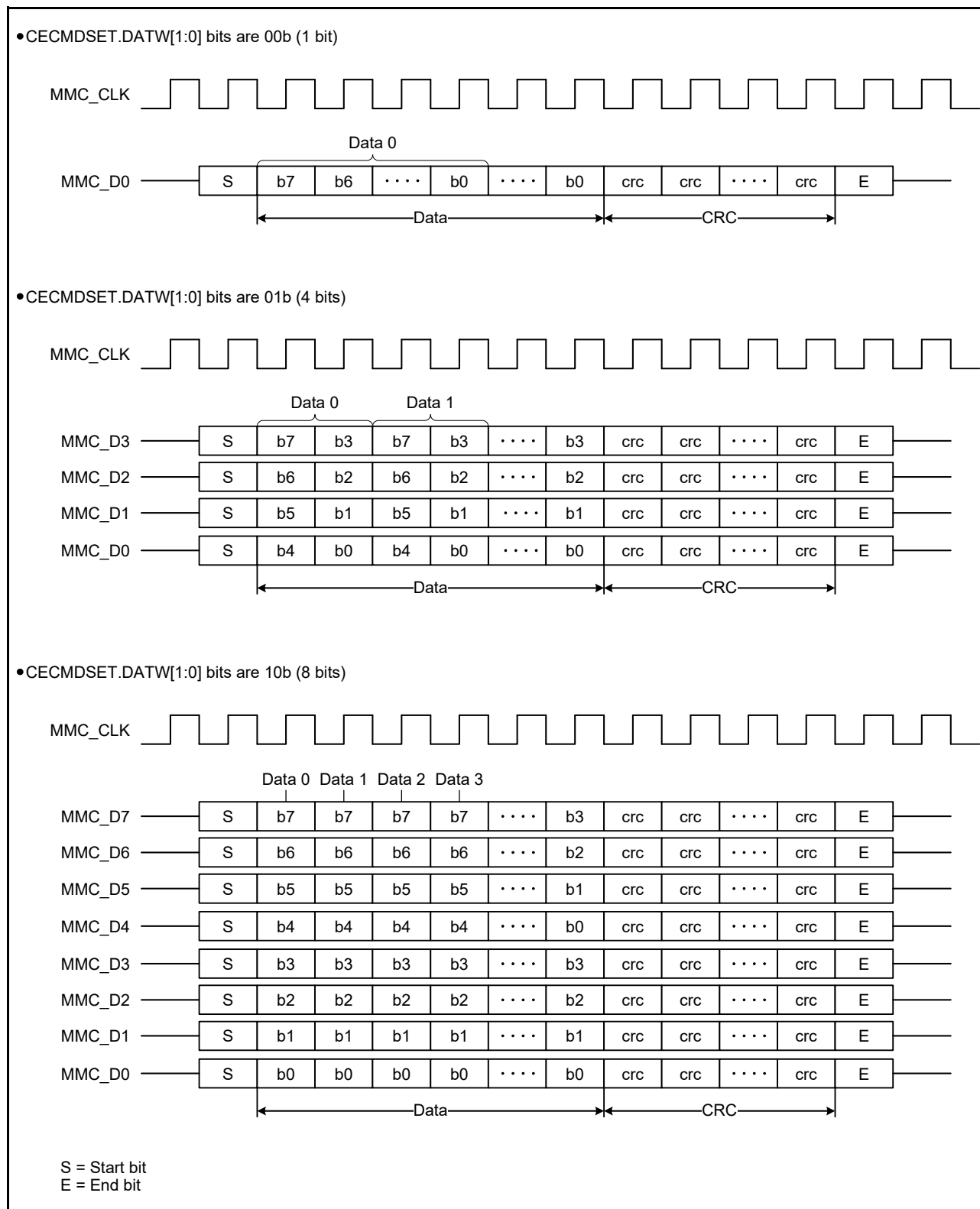


Figure 49.5 Data Block Format

49.3.3 MMCIF Buffer Structure and Access Method

Figure 49.6 shows the structure of the MMCIF buffer. The MMCIF buffer is a double buffer structure, and each buffer is 512 bytes. During a multi-block write operation, after transmitting one block of data (512 bytes) from the data stored in the MMCIF buffer, if the other buffer is full, the next block of data can be transmitted. During a multi-block read operation, even if one block of receive data (512 bytes) is stored in the MMCIF buffer, if the other MMCIF buffer is empty, the next block of receive data can be stored in the MMCIF buffer. If neither buffer is empty when performing a multi-block read operation, the MMCIF stops the MMCIF clock and pauses data reception. Then, when one of the buffers becomes empty, the MMCIF starts supplying the MMCIF clock and data reception resumes.

The CEDATA register is used to access the MMCIF buffer. If the transfer block size is set to $4n + 1$ bytes, $4n + 2$ bytes, or $4n + 3$ bytes, access the CEDATA register for $4(n + 1)$ bytes ($n = 0$ to 127).

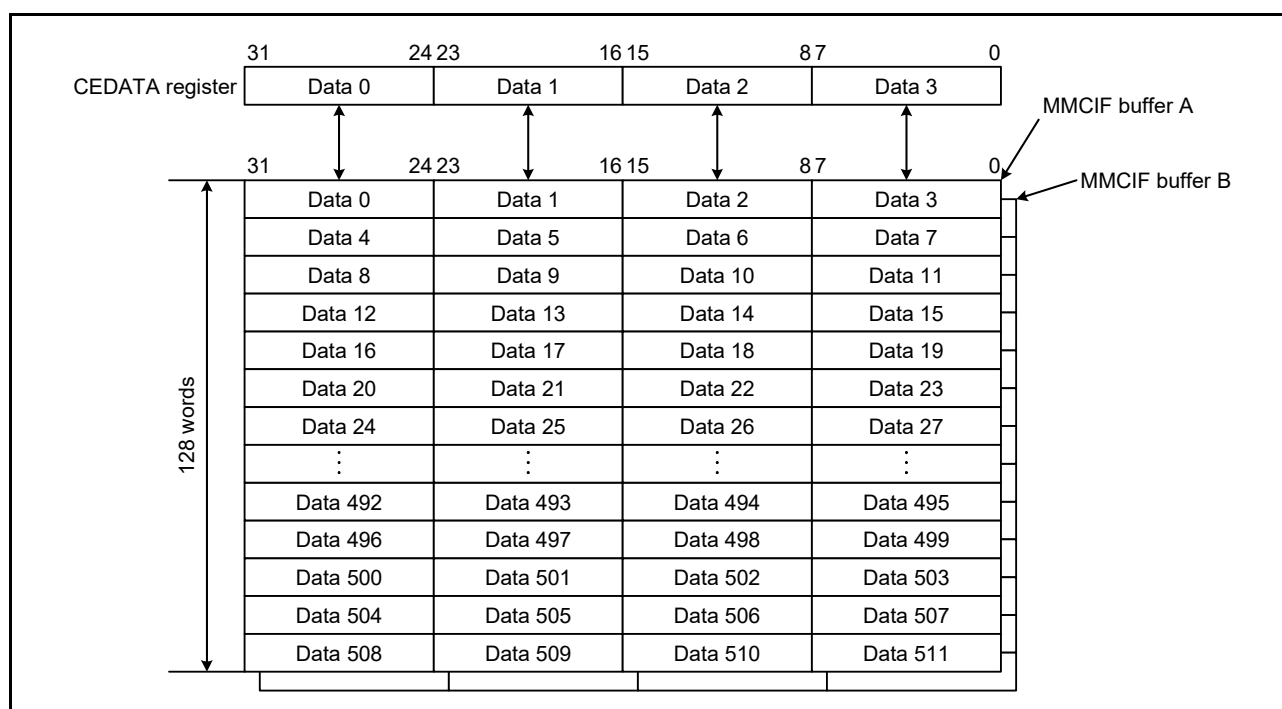


Figure 49.6 MMCIF Buffer Structure (32-Bit Access)

If the CEBUFACC.ATYP bit is set to 1, when data is written from the CEDATA register to the MMCIF buffer, or when data read from the MMCIF buffer and written to the CEDATA register, the data endian can be swapped in 1-byte units. An overview of data being swapped in 1-byte units is shown in Figure 49.7.

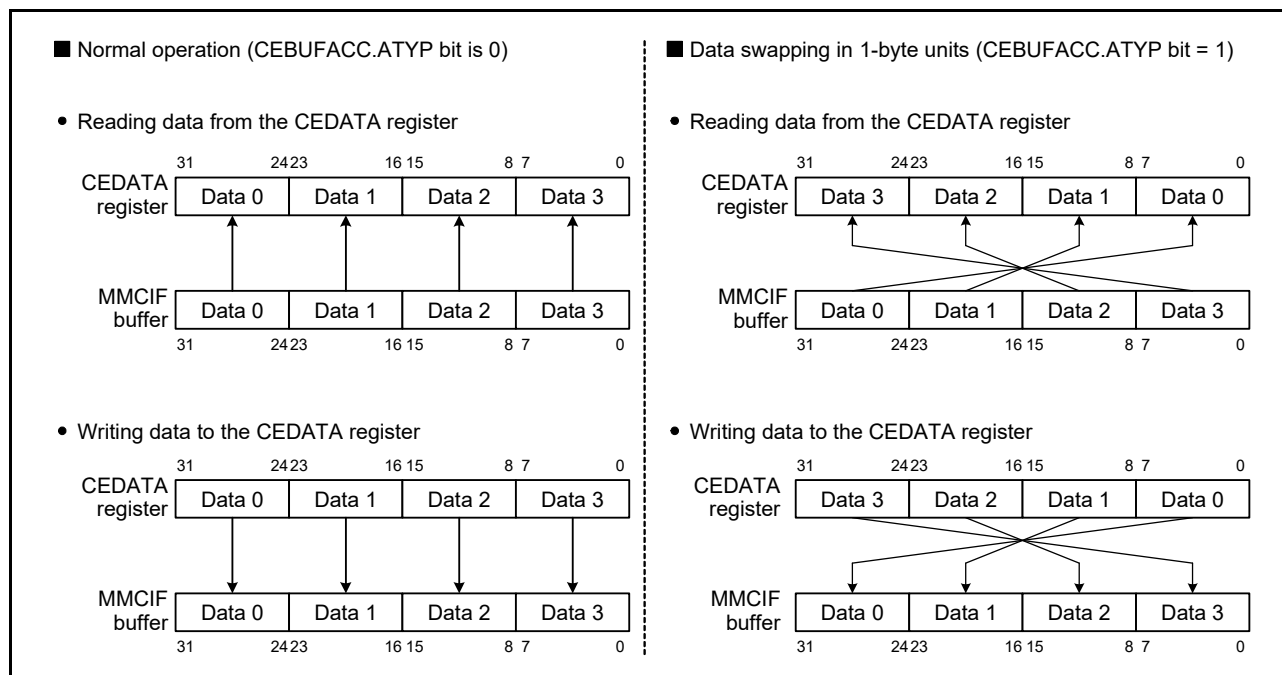


Figure 49.7 Overview of Data Swapping in 1-Byte Units (32-Bit Access)

49.3.4 Automatically Issuing CMD12

When the CECMDSET.CMD12EN bit is set to 1 and a multiple block transfer is performed, the MMCIF automatically issues CMD12. Figure 49.8 shows the Timing for Automatically Issuing CMD12 During a Multi-Block Read Operation (1-Bit Data Bus Width). CMD12 is issued such that the end bit of the command is sent 2 bits before the end bit of the data during reception of the last block.

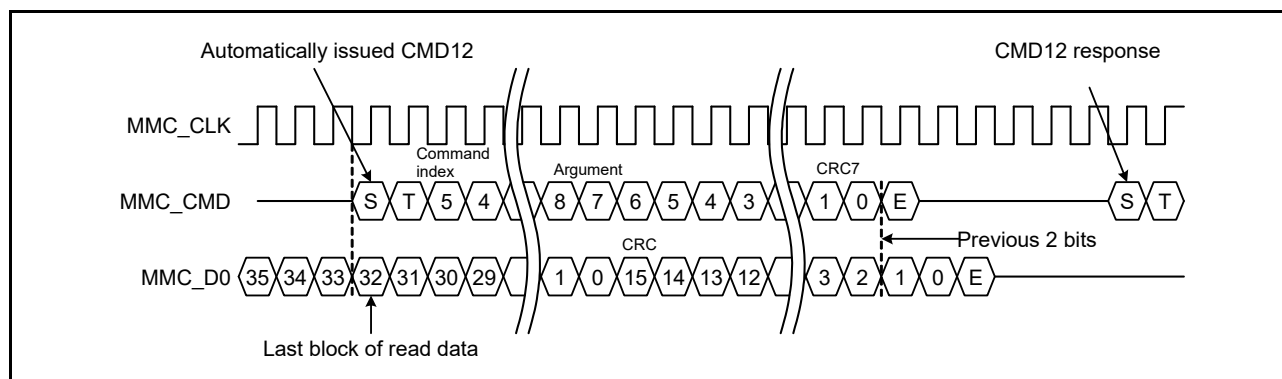


Figure 49.8 Timing for Automatically Issuing CMD12 During a Multi-Block Read Operation (1-Bit Data Bus Width)

Figure 49.9 shows the Timing for Automatically Issuing CMD12 During a Multi-Block Write Operation (1-Bit Data Bus Width). CMD12 is issued after the last block is transferred and the busy state is completed.

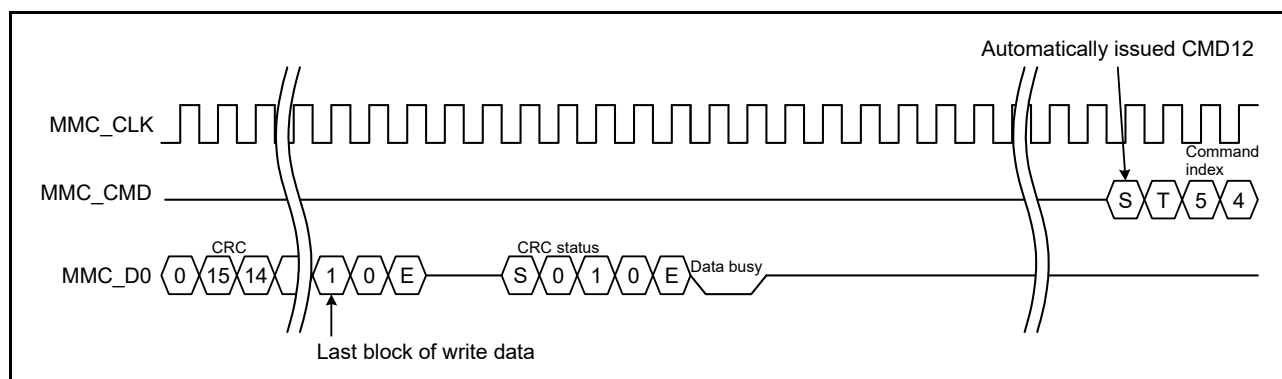


Figure 49.9 Timing for Automatically Issuing CMD12 During a Multi-Block Write Operation (1-Bit Data Bus Width)

The argument field value of the automatically issued CMD12 is set in the CEARGCMD12 register. The value for bits [39:8] of the CMD12 response are stored in the CERESPCMD12 register. Then, the response is received and the MMCIF enters the busy state.

49.3.5 MMCIF Clock Frequency During Boot Operations

Figure 49.9 shows Switching the MMCIF Clock Frequency During Boot Operations. During boot operations, when the MMC_CMD pin changes from high to low, 74 cycles of the MMCIF clock are output before the MMCIF clock frequency set in the CEBOOT.SBTCLKDIV[3:0] bits changes. After the MMC_CMD pin changes from low to high, 48 cycles of the MMCIF clock are output before the MMCIF returns to the MMCIF clock frequency set in the CECLKCTRL.CLKDIV[3:0] bits.

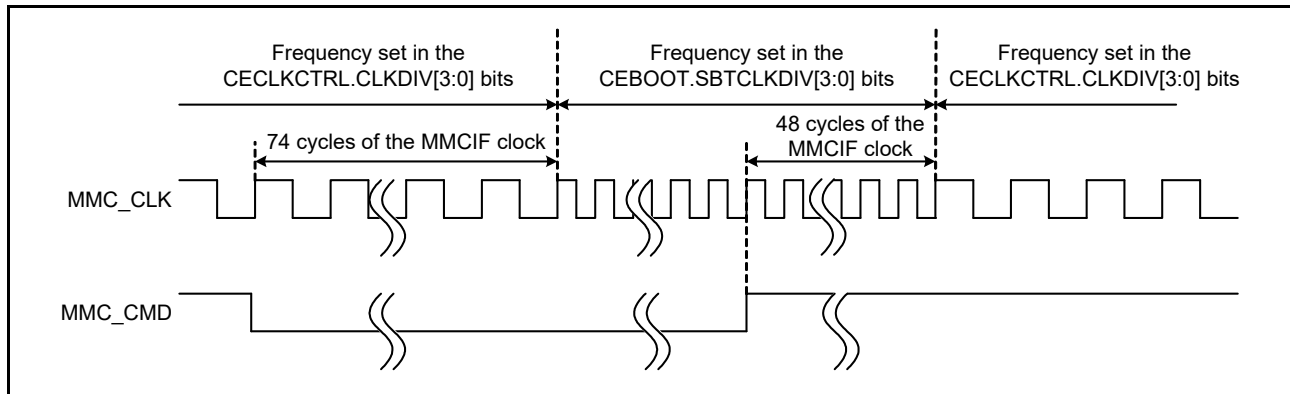


Figure 49.10 Switching the MMCIF Clock Frequency During Boot Operations

49.3.6 High Priority Interrupt (HPI)

The MMCIF supports the HPI in the sequence of CMD6, CMD24, CMD25 (pre-defined), and CMD38. The following procedures are for HPI processing.

(1) Executing an HPI while writing to the MMC

1. Forcibly stop the command sequence.
2. Wait until the CEHOSTSTS1.CMDSEQ flag becomes 0.
3. Issue CMD12 (R1) and transition the state of the MMC from rcv to prg. If the state of the MMC is already in prg at this time, the MMC will not output a response.
4. Issue CMD13 (R1).
5. Issue the HPI command (i.e. CMD12 (R1b) or CMD13 (R1b), depending on which e.MMC is connected).

(2) Executing an HPI when not writing to the MMC an while the response is busy

1. Forcibly stop the command sequence.
2. Wait until the CEHOSTSTS1.CMDSEQ bit becomes 0.
3. Issue CMD13 (R1).
4. Issue the HPI command (i.e. CMD12 (R1b) or CMD13 (R1b), depending on which e.MMC is connected).

49.3.7 Background Operations (BGOs)

To execute BGOs, issue CMD6 (R1) and write to the BKOPS_START field in the EXT_CSD register of the MMC. To stop BGOs, issue CMD6 (R1) followed by CMD13 (R1), and either confirm that the device state is in tran, or confirm that the MMC_D0 pin is high. The HPI is used to interrupt BGOs. Refer to section 49.3.6, High Priority Interrupt (HPI) for details.

49.3.8 MMCIF Processing When an Error or Timeout Occurs

When an error occurs, the MMCIF may not stop. If an error occurs during a command sequence (the CEHOSTSTS1.CMDSEQ flag is 1), after forcibly stopping the command sequence, perform a MMCIF software reset. When an error occurs, note that transmit data and receive data stored in the MMCIF buffer is not guaranteed. When a timeout occurs, the MMCIF will not stop. After a timeout occurs, before issuing the next command, forcibly stop the command sequence, perform an MMCIF software reset, and then issue the next command.

49.4 Examples of Issuing Commands

49.4.1 Issuing a Command Where a Response is Not Received and Data is Not Transferred

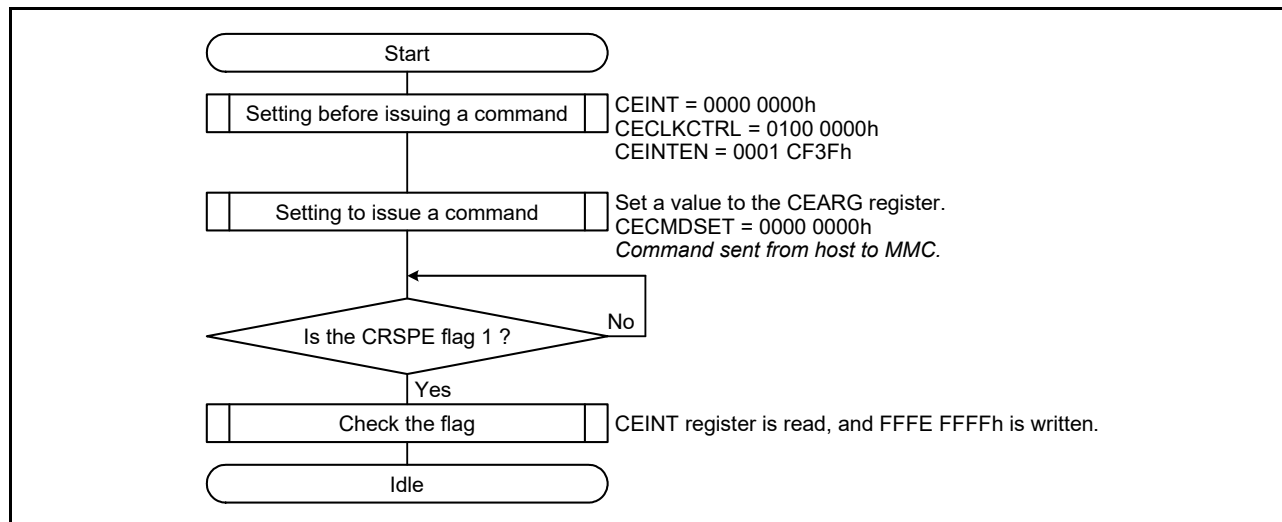


Figure 49.11 Issuing a Command Where a Response is Not Received and Data is Not Transferred (CMD0)

49.4.2 Issuing a Command Where Data is Not Transferred

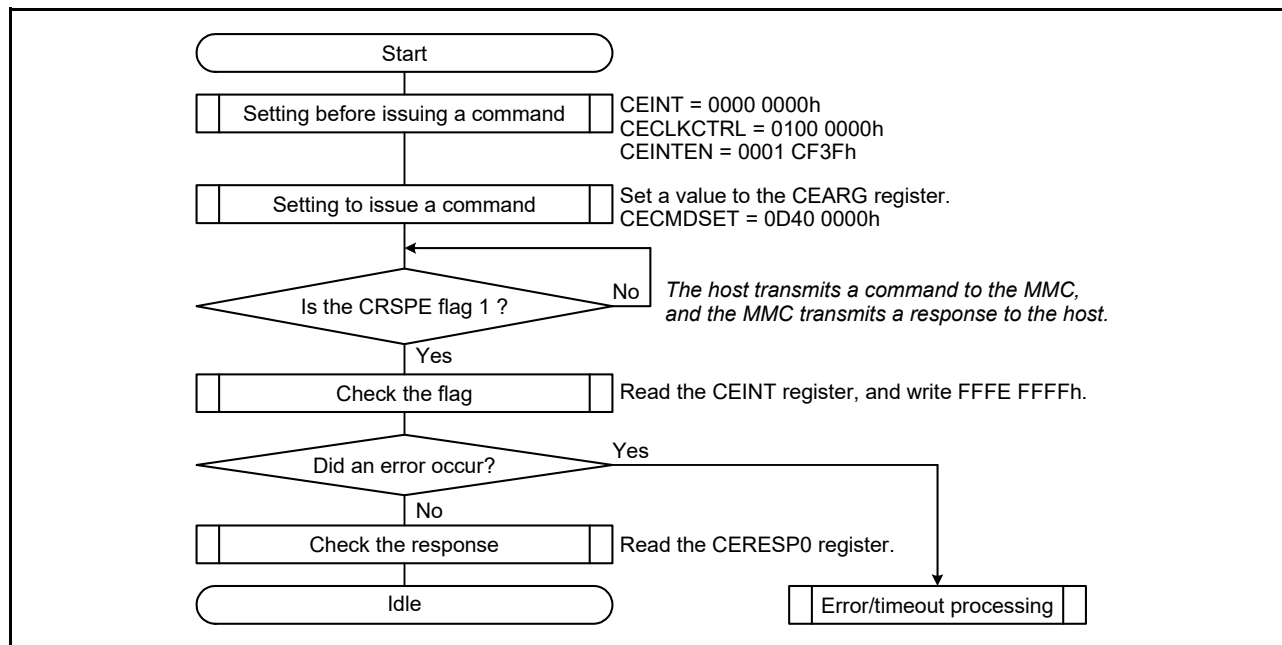


Figure 49.12 Issuing a Command Where Data is Not Transferred (CMD13)

49.4.3 Issuing a Command Where Data is Not Transferred and the Busy Signal is Used

(1) Example where the busy time is less than the period set in bits CECLKCTRL.SRBSYTO[3:0]

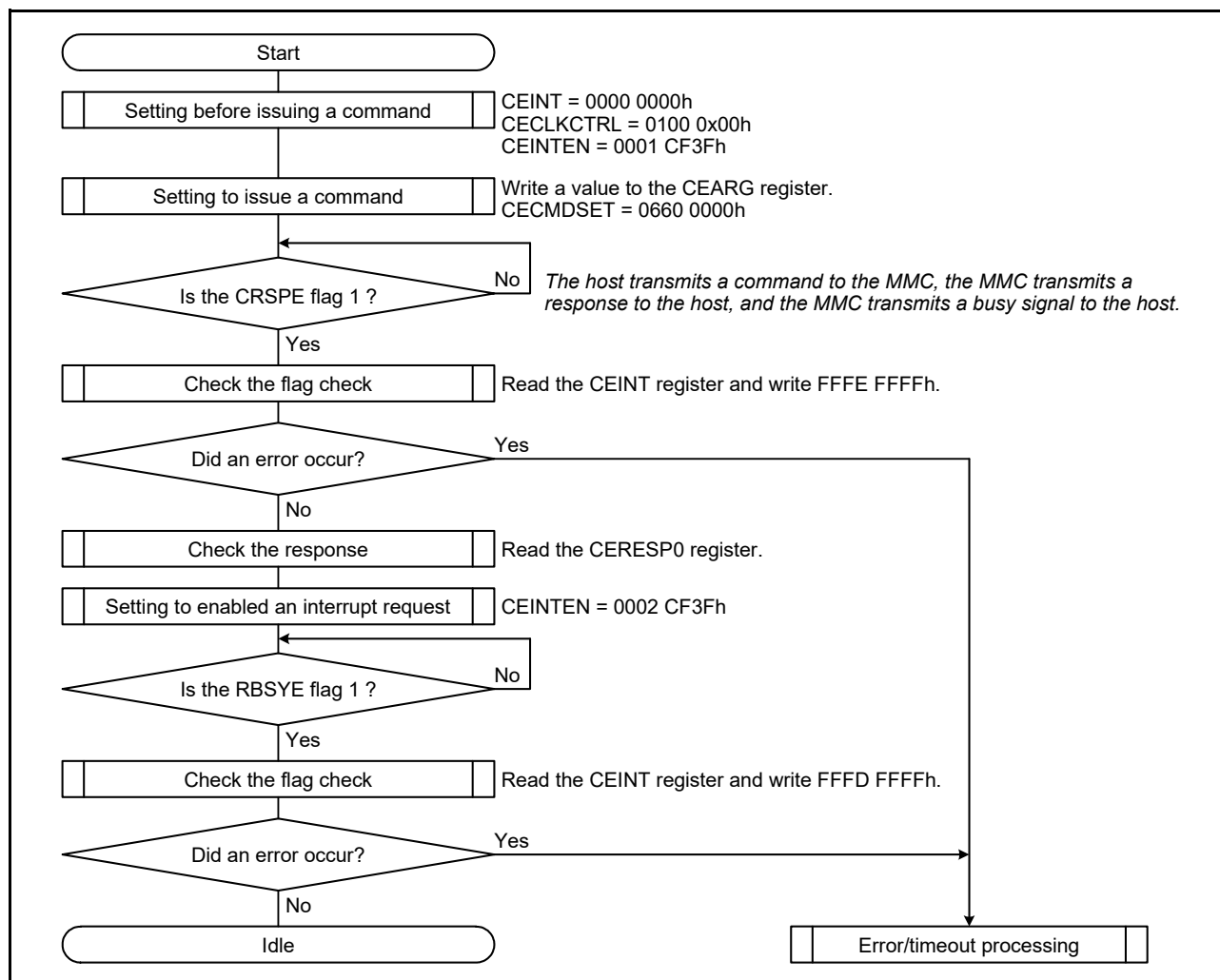


Figure 49.13 Issuing a Command Where Data is Not Transferred and the Busy Signal is Used (CMD6)

- (2) Example where the busy time is equal to or more than the period set in bits
CECLKCTRL.SRBSYTO[3:0]

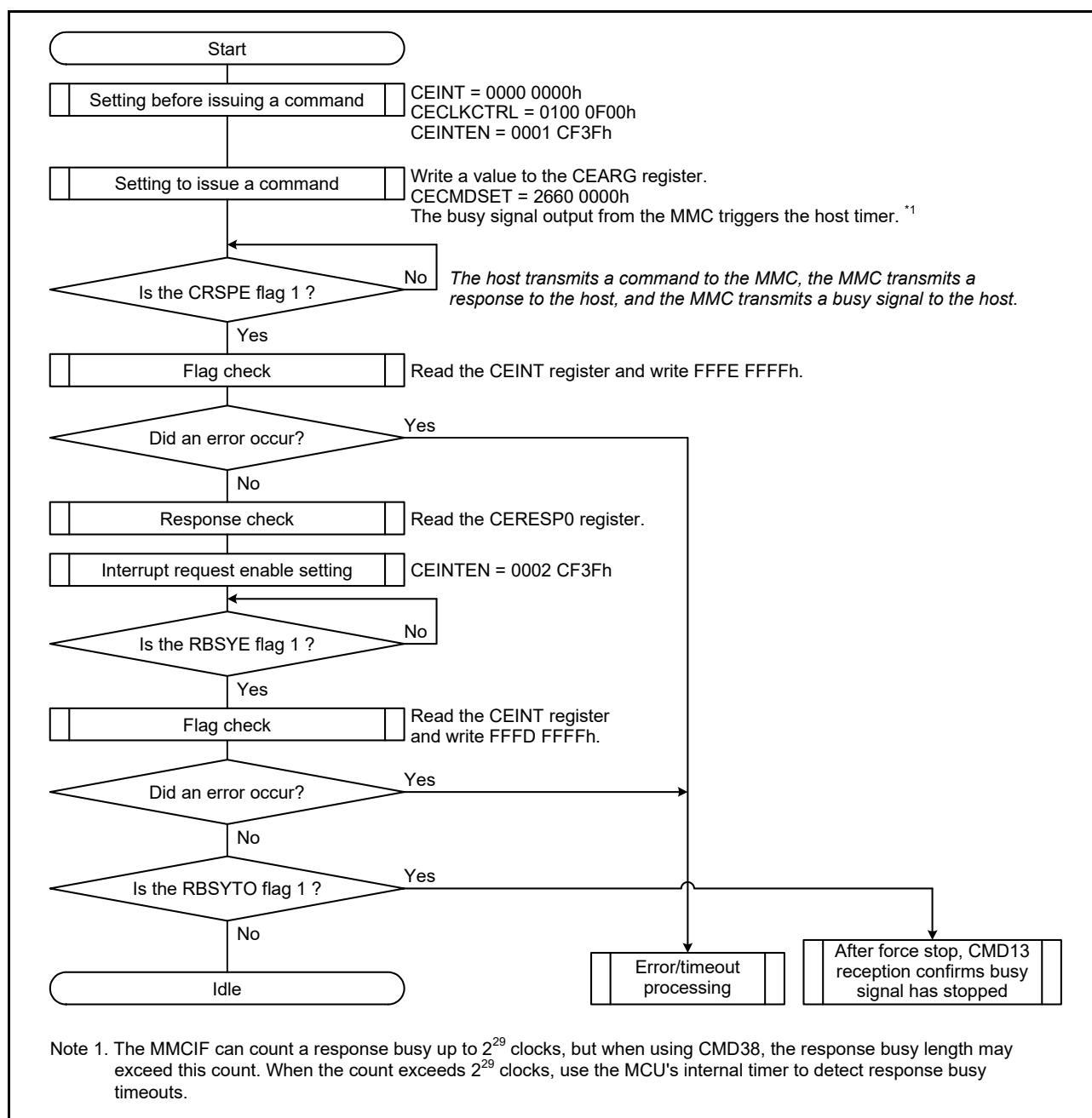


Figure 49.14 Issuing a Command Where Data is Not Transferred and Busy Signal is Used (CMD38)

49.4.4 Issuing a Single Block Read Command (CMD17)

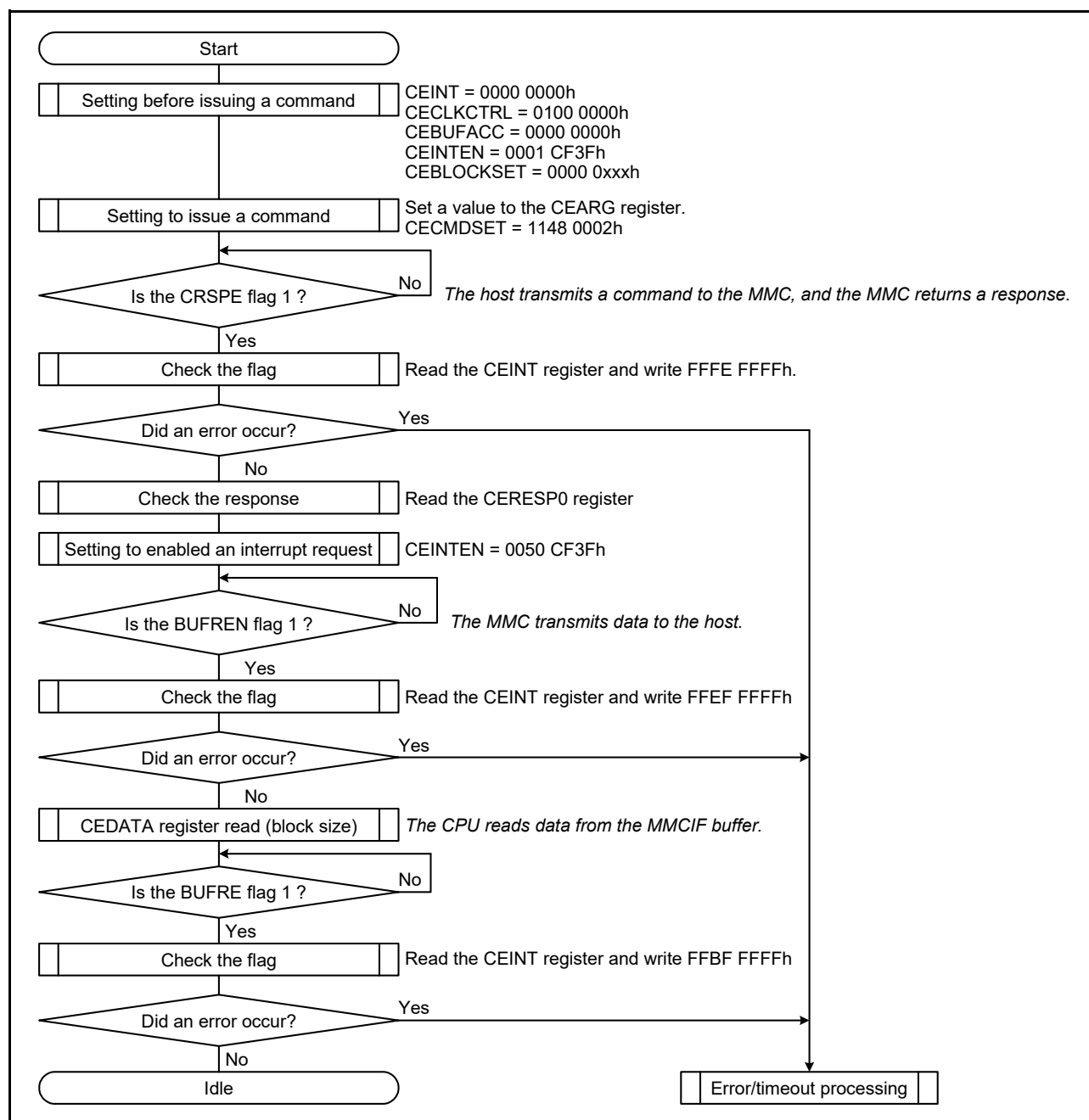


Figure 49.15 Issuing a Single Block Read Command (CMD17)

49.4.5 Issuing a Multi-Block Read Command (CMD18)

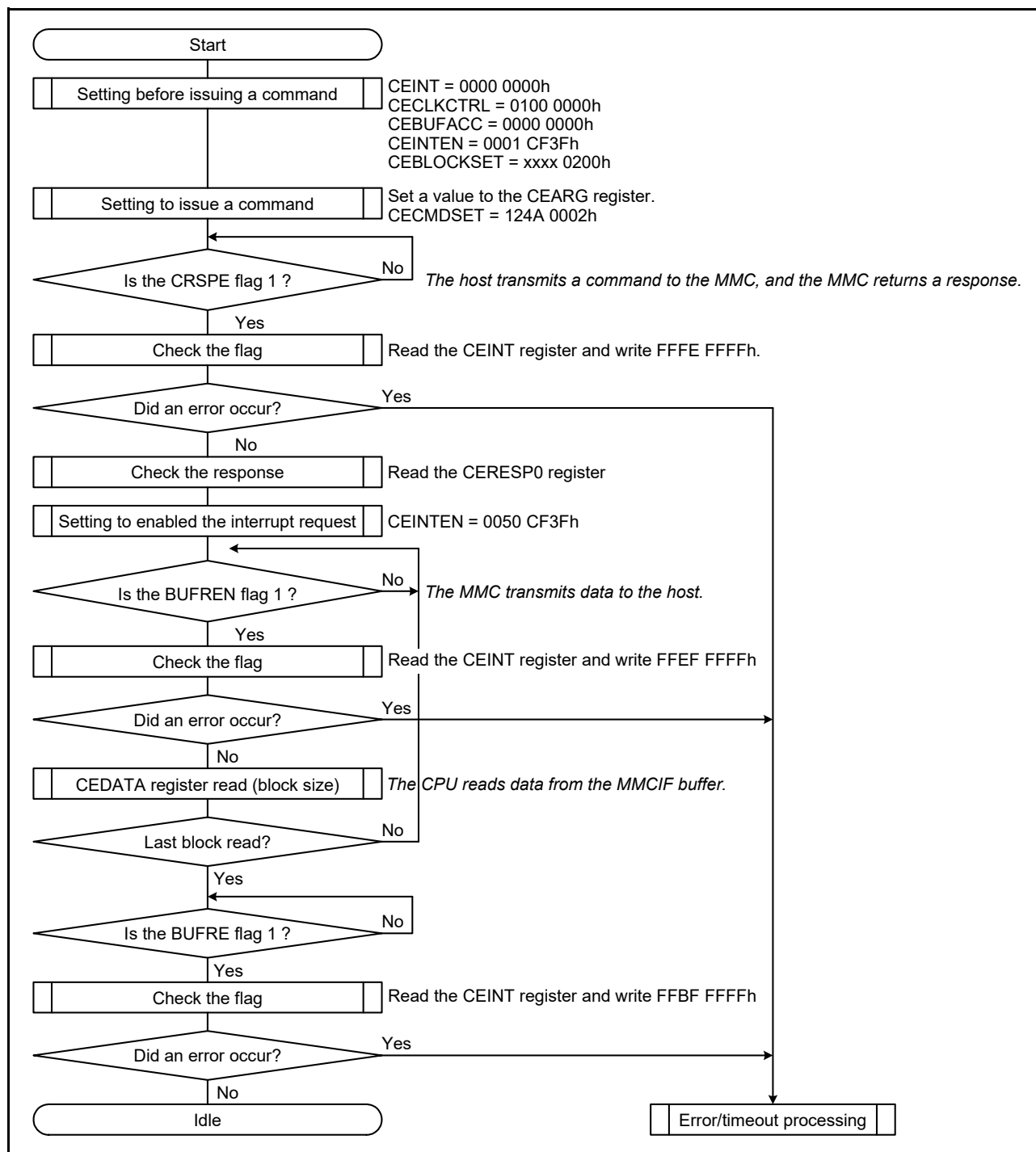


Figure 49.16 Issuing a Multi-Block Read Command (CMD18 Pre-Defined)

49.4.6 Issuing a Multi-Block Read Command (CMD18 With Automatically Issued CMD12)

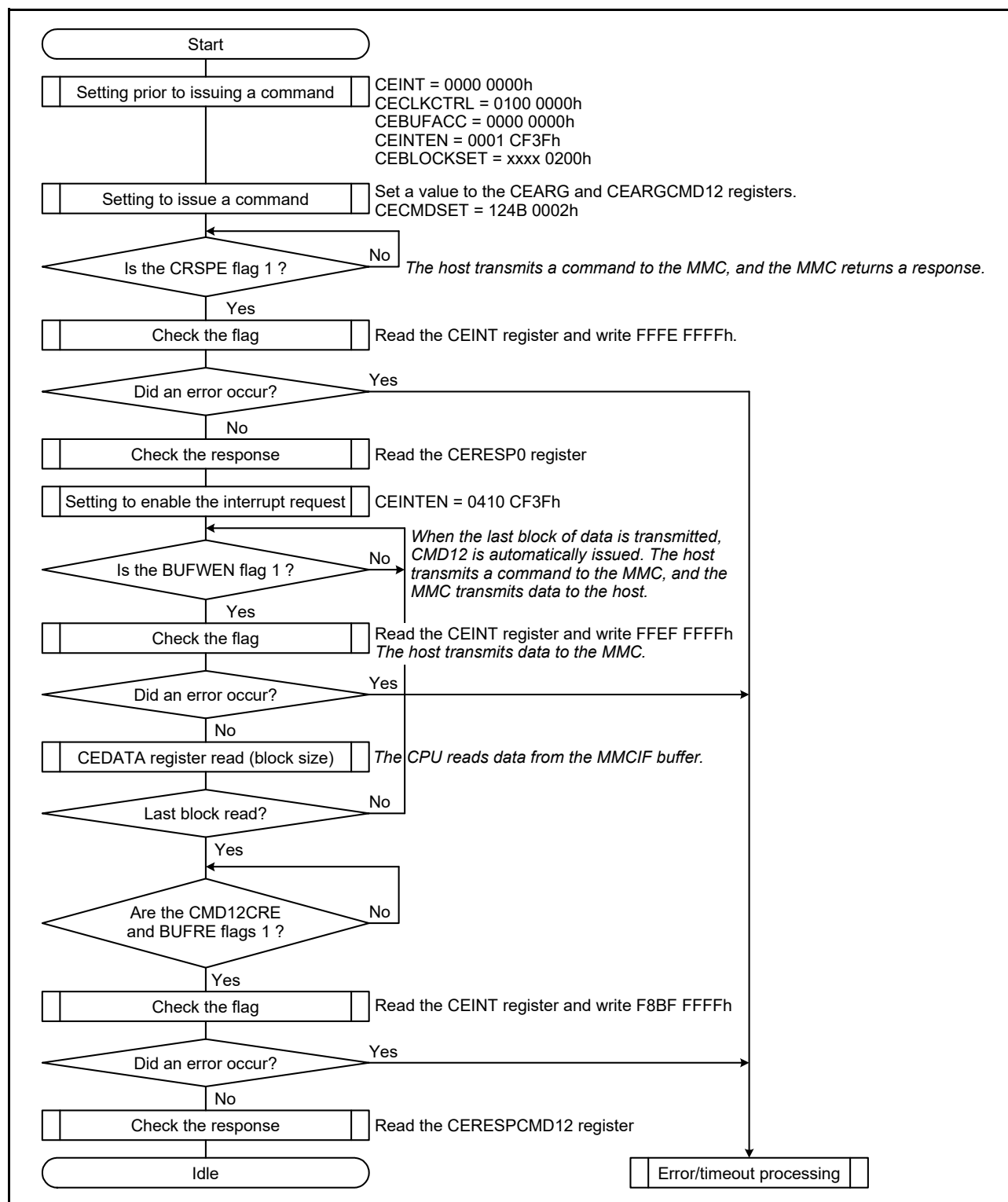


Figure 49.17 Issuing a Multi-Block Read Command (CMD18 Open-Ended)

49.4.7 Issuing a Single Block Write Command (CMD24)

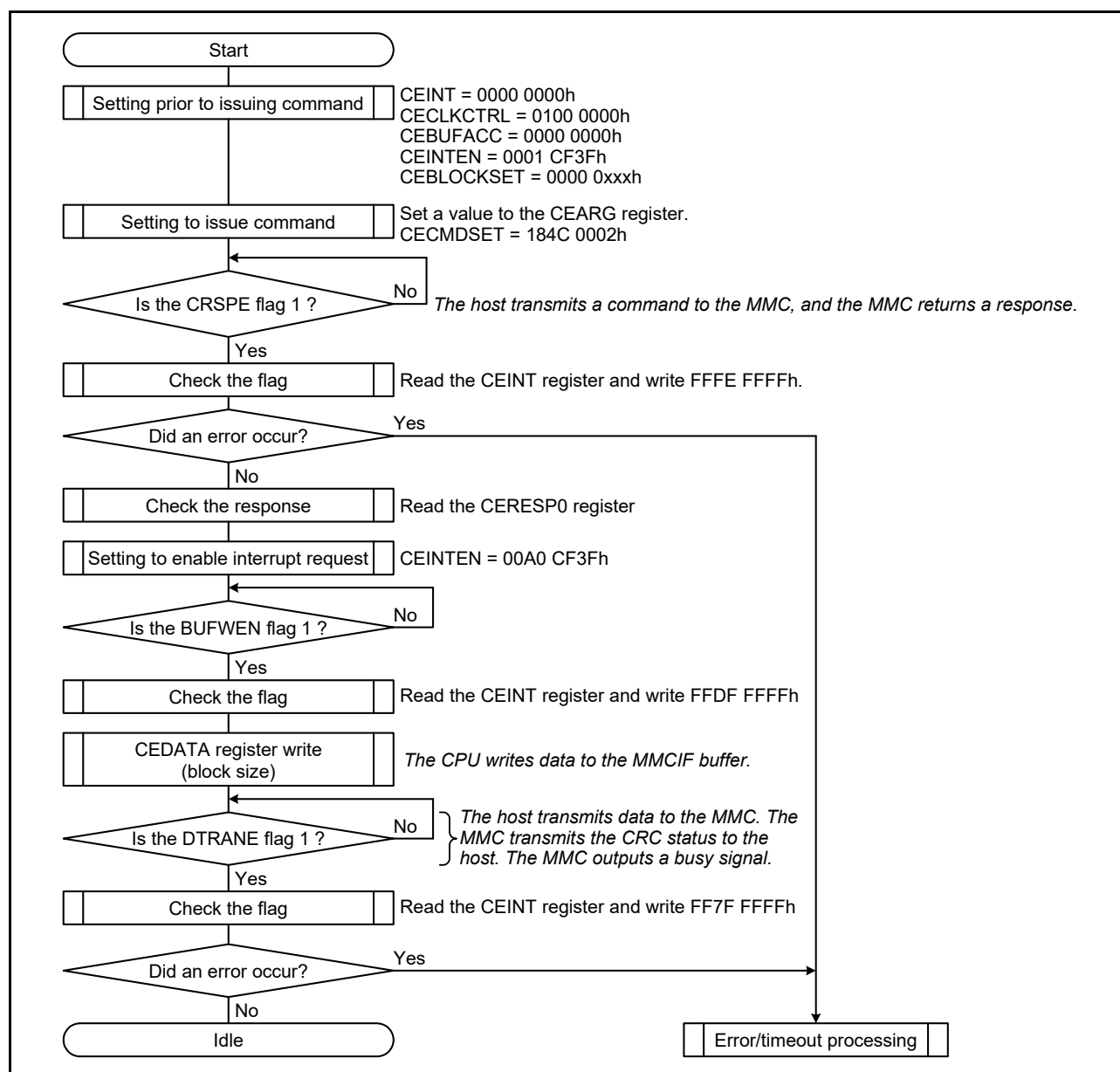


Figure 49.18 Issuing a Single Block Write Command (CMD24)

49.4.8 Issuing a Multi-Block Write Command (CMD25)

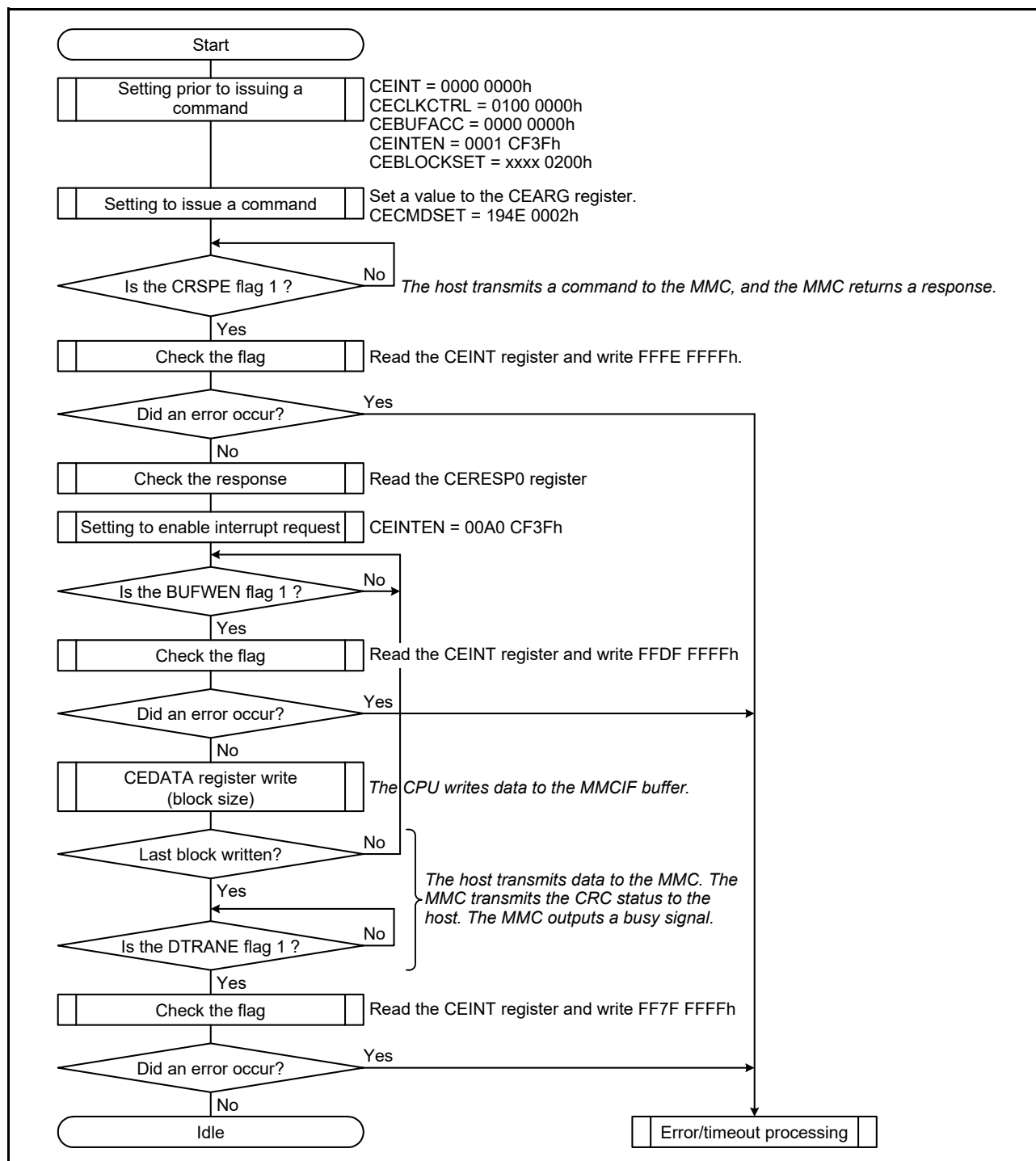


Figure 49.19 Issuing a Multi-Block Write Command (CMD25 Pre-Defined)

49.4.9 Issuing a Multi-Block Write Command (CMD25 With Automatically Issued CMD12)

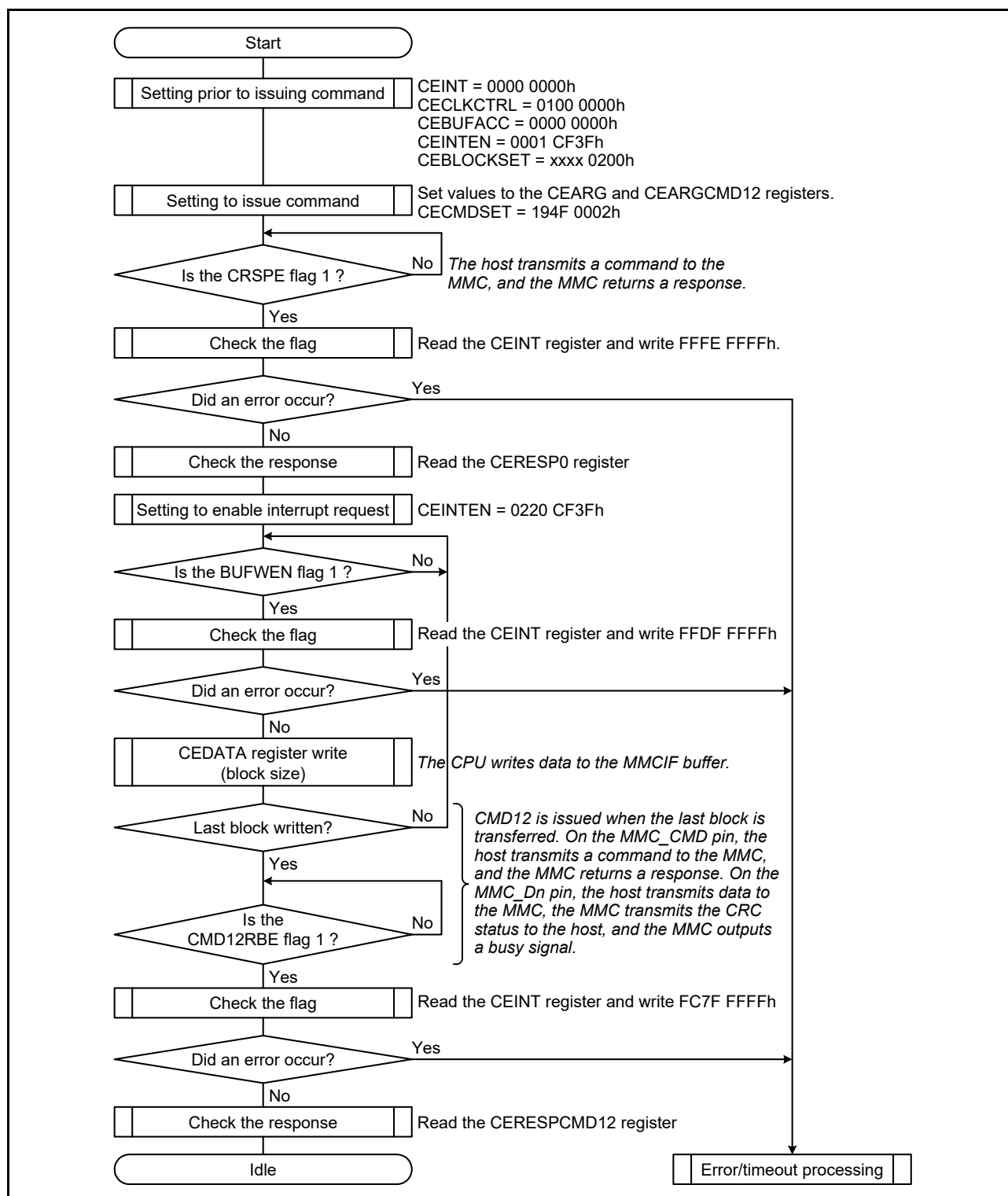


Figure 49.20 Issuing a Multi-Block Write Command (CMD25 Open-Ended)

49.4.10 Boot Operations

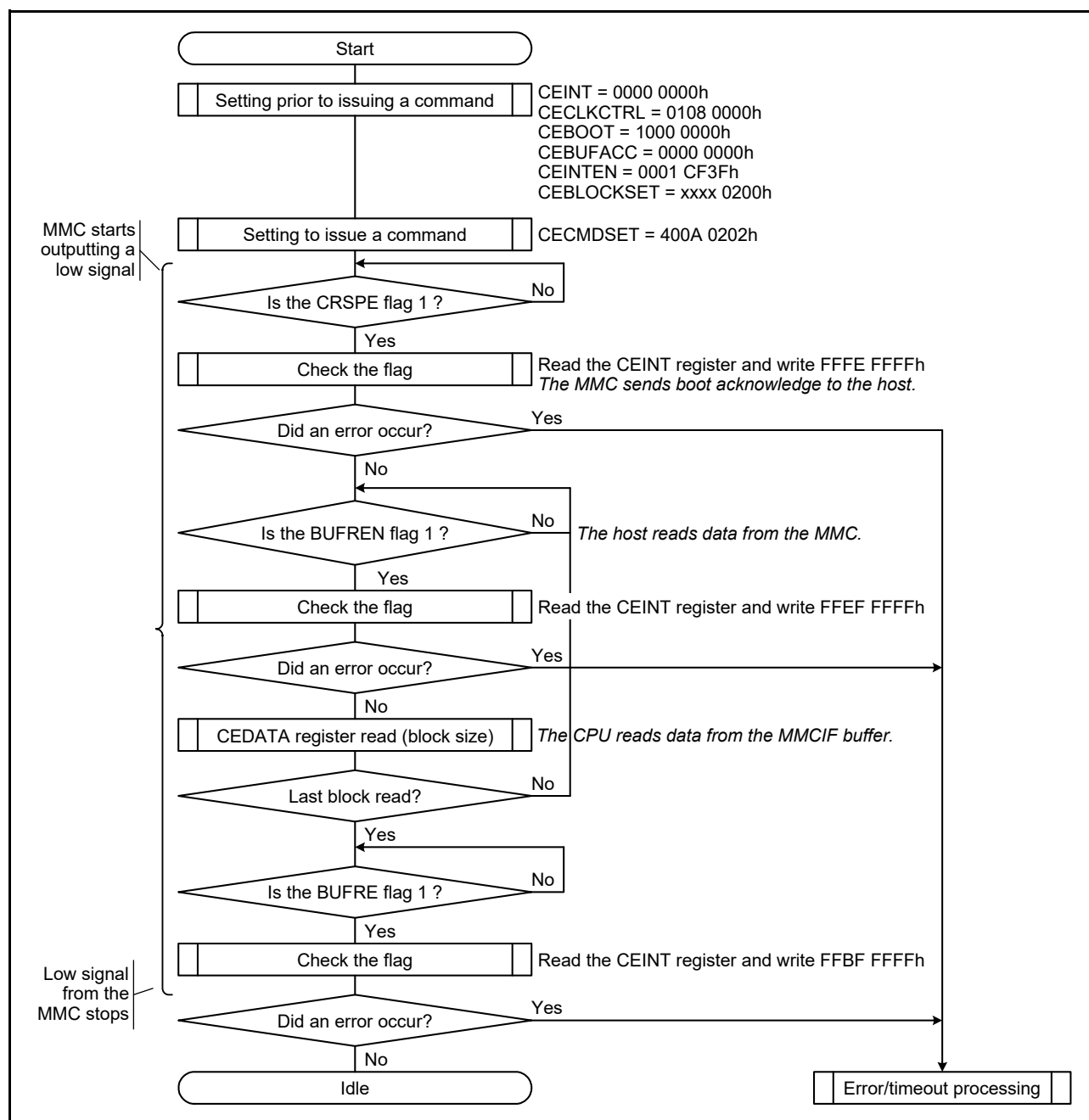


Figure 49.21 Boot Operation With Boot Acknowledge

49.4.11 Command Sequence Force Stop

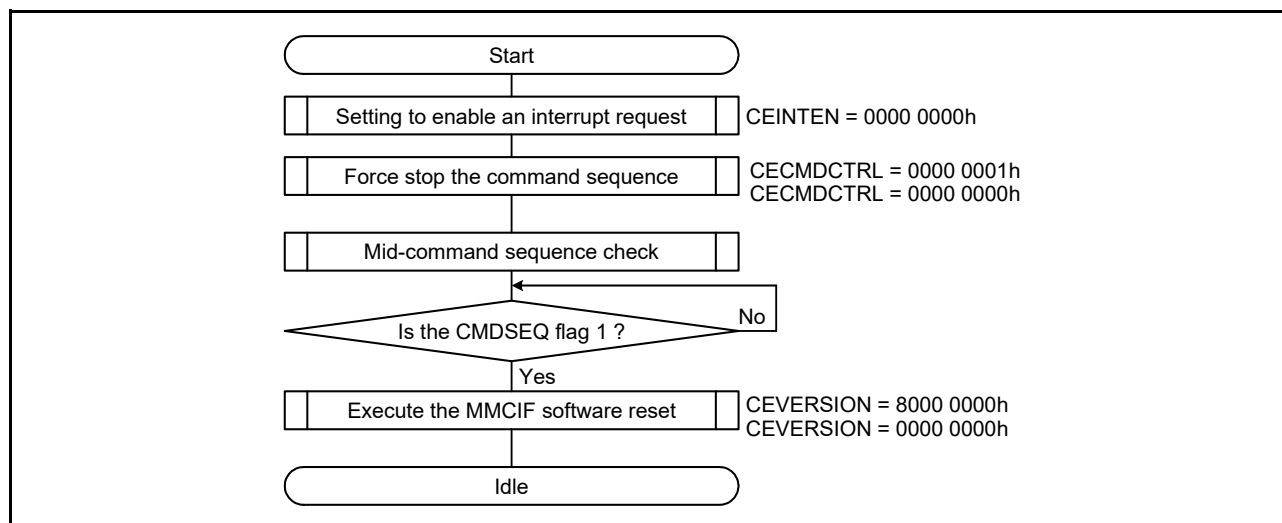


Figure 49.22 Command Sequence Force Stop

49.5 Interrupts

Table 49.5 lists the MMCIF Interrupt Sources. The MMCIF generates four types of interrupt requests: normal operation interrupt, error/timeout interrupt, MMC detection interrupt, and the MMCIF buffer access interrupt. When an interrupt status flag in the CEINT register becomes 1, if the corresponding bit in the CEINTEN register is 1, the MMCIF interrupt request is generated.

Table 49.5 MMCIF Interrupt Sources

Interrupt Source	Interrupt Status Flags		Interrupt Request Enable Register		DMAC/DTC Triggerable
	Register symbol	Bit symbol	Register symbol	Bit symbol	
Normal operation interrupt (ACCIO)	CEINT	CMD12DRE	CEINTEN	MCMD12DRE	No
		CMD12RBE		MCMD12RBE	
		CMD12CRE		MCMD12CRE	
		DTRANE		MDTRANE	
		BUFRE		MBUFRE	
		BUFWEN		MBUFWEN	
		BUFREN		MBUFREN	
		RBSYE		MRBSYE	
		CRSPE		MCRSPE	
Error/timeout interrupt (ERRIO)		CMDVIO		MCMDVIO	No
		BUFVIO		MBUFVIO	
		WDATERR		MWDATERR	
		RDATERR		MRDATERR	
		RIDXERR		MRIDXERR	
		RSPERR		MRSPERR	
		CRCSTO		MCRCSTO	
		WDATTO		MWDATTO	
		RDATTO		MRDATTO	
		RBSYTO		MRBSYTO	
MMC detection interrupt (CDETIO)	CEDETECT	CDRISE	MCDRISE	No	
		CDFALL	MCDFALL		
MMCIF buffer access interrupt (MBFAI)	—	—	CEBUFACC	DMAREN (when reading)	Yes
				DMAWEN (when writing)	

49.5.1 DMA Transfer Interrupt Requests

The MBFAI interrupt can be used to trigger DMA transfer for writing data to and reading data from the CEDATA register. When the CEBUFACC.DMAWEN bit is set to 1, if the MMCIF buffer is empty, the MBFAI interrupt request is generated. This interrupt request triggers the DMAC or DTC, and data can be transferred to the MMCIF buffer. At this point, the CEINT.BUFWEN flag does not become 1. When the CEBUFACC.DMAREN bit is set to 1, if the amount of data in the MMCIF buffer is the size specified in the CEBLOCKSET.BLKSIZ[15:0] bits, the MBFAI interrupt request is generated. This interrupt request triggers the DMAC or DTC, and data can be transferred from the MMCIF buffer. At this point, the CEINT.BUFREN flag does not become 1.

The MBFAI interrupt request is asserted for (block size set in the CEBLOCKSET.BLKSIZ[15:0] bits × number of blocks set in the CEBLOCKSET.BLKCNT[15:0] bits), and when the last block is transferred, it is negated. During DMA transfer, if an error occurs or if the command sequence is forcibly ended, the command sequence is stopped, so the MBFAI interrupt request is canceled.

Table 49.6 lists the DMAC and DTC Settings When Performing DMA Transfer.

Table 49.6 DMAC and DTC Settings When Performing DMA Transfer

Item		Settings
Transfer mode		Block transfer mode
Transfer data	1 data	32 bits
	Block size	Size set in the CEBLOCKSET.BLKSIZ[15:0] bits divided by 4
Block transfer count		Number of times set in the CEBLOCKSET.BLKCNT[15:0] bits

49.6 Notes On Using the MMCIF

49.6.1 MMC Detection

Insertion and removal of an MMC can be detected by monitoring the CEDETECT.CDRISE flag and CDFALL flag. However, as the MMCIF does not include hardware to eliminate chattering that occurs when the MMC is inserted and removed, chattering must be eliminated with software.

49.6.2 Multiple Block Transfer

Using a pre-defined multiple block transfer is recommended to increase data transfer rates.

49.6.3 Module-Stop Function Settings

Set the MSTPCRD.MSTPD21 bit to enable and disable MMCIF operation. MMCIF operation is stopped after a reset. Registers in the MMCIF can be accessed by exiting the module-stop state. Refer to section 11, Low Power Consumption for details.